

Edge-graceful labellings of odd-order trees with few vertices of degree two

Lingsen Meng

Department of Earth, Planetary, and Space Sciences,
University of California, Los Angeles, CA 90095, USA
lsmeng@g.ucla.edu

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Abstract

A tree T of odd order n is edge-graceful if its edges can be labelled bijectively by the nonzero residues modulo n so that the vertex sums are all the residues modulo n . Lee conjectured in 1989 that every tree of odd order is edge-graceful. The case of trees with at most one vertex of degree two follows from the zero-sum partition theorem of Kaplan, Lev and Roditty, and we recently settled the case of at most two such vertices. We prove that there are absolute constants C , A and n_0 such that every tree of odd order $n \geq n_0$ with D vertices of degree two is edge-graceful whenever $n > C(D+1)(\log n)^A$; in particular, for every fixed k , every large enough tree of odd order with at most k vertices of degree two is edge-graceful. The proof combines a finite table of local labelling identities, found by an exact constraint search and re-verified by an independent checker, with a bottom-up scheduling argument and with the zero-sum matching theorem of M yesser and Pokrovskiy. With the linear deletion bound for cyclic groups that its authors state in their concluding section as a consequence of their proof, the threshold becomes linear, $n > C(D+1)$. The constants are not effective. We also reduce the linear threshold to an elementary statement about permutations of an interval, which we prove in several cases and verify exactly up to order 30, and which would make the constant explicit.

1 Introduction

Let T be a tree with n vertices, n odd. An *edge-graceful* labelling of T is a bijection $\lambda: E(T) \rightarrow \mathbb{Z}_n \setminus \{0\}$ such that the induced vertex sums $s(v) = \sum_{e \ni v} \lambda(e)$, taken modulo n , form a bijection $V(T) \rightarrow \mathbb{Z}_n$. Lo [5] introduced the notion and Lee [4] conjectured that every tree of odd order is edge-graceful. The parity condition is necessary, because the sum of all edge labels is $\binom{n}{2}$, which is $\equiv 0$ modulo n only for odd n . The conjecture is surveyed in Gallian’s dynamic survey [2]. The strongest general result we are aware of is due to Kaplan, Lev and Roditty [3], who proved that $\mathbb{Z}_n \setminus \{0\}$, n odd, admits a zero-sum partition into parts of any prescribed sizes at least two, and deduced that every odd-order tree with at most one vertex of degree two is edge-graceful. Their argument identifies the obstruction exactly, since a vertex of degree two would require a zero-sum block of size one, which is impossible. In [6] we extended their theorem to trees with at most two vertices of degree two, by combining zero-sum partitions with perfect and hooked Langford sequences and a case analysis verified computationally for all odd $n \leq 5001$. The argument there is specific to the two degree-two vertices and does not extend to three.

In this paper we show that the number of degree-two vertices, rather than their placement, is what limits the zero-sum approach, provided that it is small compared with the order of the tree.

Theorem 1.1. *There are absolute constants C , A and n_0 such that every tree T of odd order $n \geq n_0$ with D vertices of degree two is edge-graceful whenever $n > C(D + 1)(\log n)^A$. In particular, for every $k \geq 0$ there is an n_k such that every tree of odd order $n \geq n_k$ with at most k vertices of degree two is edge-graceful.*

The cases $k = 0$ and $k \leq 2$ of the second statement are the theorems of [3] and [6], respectively, with $n_k = 1$ there. The constants come from an asymptotic matching theorem and are not effective, and the theorem says nothing about trees in which the degree-two vertices have positive density, e.g. caterpillars with many subdivided edges, the regime of the full conjecture.

The only external input is a zero-sum matching theorem in $\mathbb{Z}_n$ that tolerates a set of deleted residues, and it is the size of that set, not the matching itself, that determines the threshold. The printed statements of M yesser and Pokrovskiy [7] tolerate $n/(\log n)^A$ deletions and give Theorem 1.1. For cyclic groups, however, the authors state in their concluding Section 7 that a linear number of deletions can be tolerated: the only source of a logarithm in their proof is a bound on the length of a commutator product, which is empty in an abelian group, and they record the consequence $g(n) = \Omega(n)$ for $\mathbb{Z}_n$ in their Problem 7.4. With that bound our argument gives a linear threshold.

Theorem 1.2. *There are absolute constants C and n_0 such that every tree T of odd order $n \geq n_0$ with D vertices of degree two is edge-graceful whenever $n > C(D + 1)$. In particular, for every $k \geq 0$, every odd-order tree with at most k vertices of degree two and more than $C(k + 1)$ vertices is edge-graceful, since $D \leq k$.*

The proof of Theorem 1.2 uses the zero-sum matching lemmas of [7] in the linear form for cyclic groups that the authors state in their Section 7; that form is not a numbered theorem there, and Remark 6.2 records exactly what is used and why it follows from their proof. The qualitative statement, that a bounded number of degree-two vertices is no obstruction once the tree is large, does not depend on it at all. In Section 7 we reduce the linear threshold, with an explicit constant and without any use of [7], to a conjecture on permutations of an interval that we prove in several cases and verify exactly up to order 30.

Our approach separates the problem into a local part and a global part. Locally, each maximal path of degree-two vertices, together with the branch vertex that owns it, is labelled by an explicit family of integer forms whose vertex sums permute its own labels (we call this the $P=O$ property). Such families are found by our exact search over all output permutations and constraint programming, and every family used in the proof is re-verified by a program that shares no code with the search. Globally, the families are assembled bottom-up so that the multiset of special labels is closed under negation up to at most one zero-sum triple and has size and magnitude linear in D . The remaining labels are then partitioned into zero-sum blocks of the prescribed sizes by the random Hall–Paige theorem of M yesser and Pokrovskiy [7]. Therefore the local part is a finite computation (12,748 certified families), whereas the global part is a scheduling lemma, whose regression is an executable scheduler that we tested on all trees with at most nine vertices and on thousands of random trees.

The paper is organised as follows. Section 2 reformulates edge-gracefulness as a permutation identity and fixes the vocabulary. Section 3 decomposes the tree and proves the counting lemmas. Section 4 describes the local families and the finite certificate table. Section 5 proves the sequential realization lemma. Section 6 applies the dense completion and proves Theorems 1.1 and 1.2. Section 7 records the elementary route to the linear threshold. Section 8 describes the computations and the certificate format.

2 Set-up

Throughout, T is a tree of odd order n rooted at a vertex ρ of degree at least three (if T is a path the explicit labelling $1, 2, \dots, n-1$ along the path is edge-graceful, so this case is excluded). For $v \neq \rho$ let e_v denote the parent edge of v and put $\lambda(e_\rho) = 0$. A vertex is *stationary* if the labels of its child edges sum to zero, so that $s(v) = \lambda(e_v)$. Let W be the set of non-stationary vertices together with ρ and let $S = \{\lambda(e_w) : w \in W \setminus \{\rho\}\}$, the *special support*.

Lemma 2.1. λ is edge-graceful if and only if the multiset $\{s(w) : w \in W\}$ equals $\{0\} \cup S$.

Proof. For a stationary vertex $v \neq \rho$ the sum $s(v)$ equals the label of e_v , so the stationary vertices contribute exactly the labels of their parent edges. The remaining labels are $\{0\} \cup S$ and they must be produced by W . $\square$

Every vertex of degree two is in W , because a single nonzero child label cannot sum to zero. Therefore our construction chooses W of size $O(D)$, arranges the identity of Lemma 2.1 cell by cell, and leaves every other internal vertex a zero-sum block of size zero or at least two. We work with integer labels of magnitude $O(D)$ and reduce modulo n at the end.

Core and arms. Suppressing the degree-two vertices of T gives the core Q , a tree whose vertices have degree one or at least three in T , and a core edge carries a length ℓ , the number of suppressed vertices. We process Q bottom-up. A subdivided child edge of a core vertex v is an *arm* of v when its lower endpoint is not an owner (defined below), and the *head chain* of the lower endpoint otherwise. A direct child edge whose lower endpoint is not an owner is a *port*.

Cells and $P=O$. A *cell* at a core vertex v consists of the labels of v 's parent edge (or of its head chain), of its residual arms and of some recruited ports, and possibly of the edge to one or two active children. Its *physical multiset* P is the multiset of these labels. Its *output multiset* O consists of the sum at v , the adjacent sums along each arm and chain, the terminal label of each arm, and the label of each port. A cell is *self-permuting*, written $P=O$, when the two multisets coincide, and for a root cell the constant 0 is adjoined to P . Substituting a $P=O$ cell for a stationary child edge preserves the identity of Lemma 2.1 because the shared edge is counted once on either side.

3 Decomposition and counting

3.1 Owner-free neutralization

Lemma 3.1 (neutral units). *Two arms whose lengths have the same parity and whose residues modulo six are not $\{1, 3\}$, and the four-arm multisets with residues $(1, 1, 1, 3)$ and $(1, 3, 3, 3)$, admit labellings whose adjacent sums and terminals permute their own labels, whose first labels sum to zero, and whose support is closed under negation, and each has two free integer parameters. Two arms of lengths one and three admit no such labelling.*

The thirteen families for the residue representatives are found by our constraint search of Section 4 (a root row with pinned zero output) and re-verified by the independent checker. They extend to all lengths in the residue classes by the six-edge insertion, which lengthens an arm by six while adding three inverse pairs, and the impossibility for lengths one and three is an exact enumeration. We call these labelled arm sets *neutral units*. (Pairs whose residues are $\{1, 3\}$ but whose lengths are not, e.g. $(1, 9)$ or $(3, 7)$, also admit such labellings by the same

search. We do not use them, since a pair of this type is carried instead in the residual, which has families of its own.)

Corollary 3.2 (six residual types). *The arm multiset of every core vertex can be partitioned into neutral units and a residual of one of six types, namely empty, one odd arm, one even arm, a pair with residues $\{1, 3\}$, one odd and one even arm, or such a pair together with one even arm.*

Proof. Pair the even arms arbitrarily, leaving at most one even arm. Among the odd arms, pair every arm of residue five with any other odd arm, then pair arms of residue one with each other and arms of residue three with each other. What remains is at most one arm of residue one and at most one of residue three. Thus the odd part of the residual is empty, a single odd arm, or a pair with residues $\{1, 3\}$, and the even part is empty or a single arm. $\square$

3.2 Owners and contexts

Definition 3.3. A nonroot core vertex is an *owner* if its residual is nonempty, or if exactly one active child remains after the cancellations described below and no port is available to absorb it. A vertex with empty residual and no remaining active child is stationary, and its parent edge, if subdivided, is an arm of its parent.

At a vertex with $k \geq 2$ active children (children which are owners) the heads are free parameters of the children's cells, so they are prescribed to sum to zero (a mirror pair for $k = 2$, a free zero-sum triple for $k = 3$, pairs and at most one triple otherwise), and none remains. A mirror pair is closed under negation, but a cancelling triple is not, and its negatives are placed at a sink in the ordinary part of the tree exactly as the negatives of a token are; Appendix B counts these triples together with the tokens. A single remaining active child with head x is absorbed by a stationary vertex with q ports by a helper port labelled $-x$ ($q = 1$ or $q \geq 3$) or by two ports labelled a and $-x - a$ ($q = 2$). Two exceptional child types cannot be absorbed in this way, because every $P=O$ family of the child contains the antipode of its own head: the residue-one letter and the (1; one port) bottom, and such children are joined into their parent's cell.

The *context* of an owner is the tuple (h, R, p, c) where h is the length of its head chain reduced modulo six, R its residual arm multiset with lengths reduced modulo six, $p \in \{0, 1, 2, 3\}$ the number of recruited ports (never leaving exactly one unrecruited), and c the child type, one of *none*, *free*, the two joined types, or the two-input type used when two active heads are kept. The root has context (root, R, p, c) .

Reduction modulo six is justified by the following identity, which lengthens one arm of a cell by six and changes nothing else.

Lemma 3.4 (six-edge insertion). *Let an arm of a cell carry the labels $a_0, \dots, a_\ell$, a_0 at the owner, so that it contributes the outputs $a_0 + a_1, \dots, a_{\ell-1} + a_\ell$ and a_ℓ . Write $c = a_0$ and $d = a_1$ and insert the six labels*

$$\frac{-2c-d}{3}, \quad \frac{c-d}{3}, \quad \frac{c+2d}{3}, \quad \frac{2c+d}{3}, \quad \frac{-c+d}{3}, \quad \frac{-c-2d}{3}$$

between a_0 and a_1 . They form three pairs of opposite forms, and the seven outputs they create are exactly those six labels together with the output $c + d$ that the arm had before. For a port, an arm of length zero whose only label is c and whose output is c itself, the same holds with the six labels

$$\frac{c}{3}, \quad \frac{-2c}{3}, \quad \frac{4c}{3}, \quad \frac{-c}{3}, \quad \frac{2c}{3}, \quad \frac{-4c}{3}$$

and with c in place of $c + d$.

Proof. For the first list the outputs are, in order, $c + \frac{-2c-d}{3} = \frac{c-d}{3}$, $\frac{-2c-d}{3} + \frac{c-d}{3} = \frac{-c-2d}{3}$, $\frac{c-d}{3} + \frac{c+2d}{3} = \frac{2c+d}{3}$, $\frac{c+2d}{3} + \frac{2c+d}{3} = c + d$, $\frac{2c+d}{3} + \frac{-c+d}{3} = \frac{c+2d}{3}$, $\frac{-c+d}{3} + \frac{-c-2d}{3} = \frac{-2c-d}{3}$ and $\frac{-2c-d}{3} + d = \frac{-c+d}{3}$, which is the asserted multiset, and the three pairs are the first with the fourth, the second with the fifth and the third with the sixth. The port list is checked the same way. $\square$

Thus a family for a context is also a family for the context with any one arm six longer, and in particular a port may be replaced by an arm of length six. The new labels are forms in the two labels the insertion sits between, so the identity is one statement about all cells at once. The insertion may make two labels of the cell equal for special values of c and d , which the search rules out by requiring the label forms to stay pairwise distinct. A twelve-label endpoint preserving insertion does the same for head chains.

A second identity removes a much larger part of the search. Most of the contexts of the list differ from a narrower one only by a set of arms and ports that can be made to pay for itself, and such a set may be adjoined to any family at all.

Definition 3.5. A *block* is a multiset of arm lengths $\ell_1, \dots, \ell_k$ together with a number $m \geq 0$ of ports, and it carries $N = m + \sum_i (\ell_i + 1)$ labels. The block is *neutral* if there are integers $c_1, \dots, c_N$ such that, when its labels are taken to be $c_1 t, \dots, c_N t$ in a single parameter t ,

- (N1) the labels it contributes to the owner, namely its m port labels and the first label of each of its arms, sum to zero;
- (N2) the multiset of its N labels equals the multiset of the N outputs at the vertices it introduces, that is the m port leaves and, for an arm $a_0, \dots, a_\ell$, the outputs $a_0 + a_1, \dots, a_{\ell-1} + a_\ell$ and a_ℓ ;
- (N3) the multiset $\{c_1, \dots, c_N\}$ is closed under negation; and
- (N4) the c_i are nonzero and pairwise distinct.

It is a *token block* if it satisfies (N1), (N2) and (N4) and, instead of (N3), three of its labels sum to zero and span a plane while the remaining $N - 3$ are closed under negation. A token block therefore needs at least two parameters, and N is odd for it and even for a neutral block.

Which of the two a given set of arms and ports can be is decided by parity alone. Two arms of lengths a and b carry $a + b + 2$ labels, so a same-parity pair can only be a neutral block and a pair of opposite parity can only be a token block, which is the arithmetic behind the neutral units of Section 3: an $\{o, e\}$ residual is exactly the residual that forces a token.

Lemma 3.6 (neutral adjunction). *Let B be a neutral block. Then every family for a context C is also a family for the context obtained by adjoining the arms and the ports of B to the owner of C , with the same head, the same pinned input and the same token. If B is a token block the same holds provided C has at most one token, and the enlarged family carries the triple of B as its token when C has none and as its second token when C has one.*

Proof. Give B a parameter t that the family for C does not use. Every label of C is then zero in t and, by (N4), no label of B is, so the labels of the enlarged cell are nonzero, they are pairwise distinct across the two groups because they differ in the coordinate t , and they are pairwise distinct inside B again by (N4). By (N1) the owner keeps the output it had, and the only other outputs that change are those created inside B , so $P = O$ for the enlarged cell follows from $P = O$ for C together with (N2). Condition (N3) splits the new labels into pairs of opposite forms, which extends the signature partition by mirror pairs and leaves the token where it was. For a token block the same partition is extended by the mirror pairs together

with the new triple, which sums to zero and spans a plane as the signature demands, and which is disjoint from the token of C because it lives in the fresh parameters. Finally the head chain, the pinned input, the token and the ports of C are untouched, so the free-head condition, the input condition, the token rank and each of the variant conditions of Section 4 hold for the enlarged family exactly as they held before. $\square$

The two blocks we use most often are neutral for a reason one can write down.

Lemma 3.7. *Two arms of the same length ℓ form a neutral block, and so do two ports. In addition, two arms of lengths $a < b \leq 8$ of the same parity form a neutral block for every such pair except $\{1, 3\}$, which is not neutral; one port together with an arm of even length $\ell \in \{4, 6, 8\}$ is neutral; one port together with two arms of opposite parity is neutral; and two ports together with an arm of odd length $\ell \in \{5, 7\}$ are neutral. In addition, two arms of opposite parity $\{o, e\}$ form a token block for every pair with $o \in \{3, 5\}$, $e \in \{2, 4, 6\}$ and for $\{1, 6\}$ and $\{2, 5\}$, although $\{1, 2\}$ and $\{1, 4\}$ do not.*

Proof. For two arms of length ℓ take $a_i = (-2)^i t$ on one and $a_i = -(-2)^i t$ on the other. The outputs of the first are $a_i + a_{i+1} = -a_i$ for $i < \ell$ together with a_ℓ , and those of the second are their negatives, so the $2\ell + 2$ outputs are exactly the $2\ell + 2$ labels $\pm(-2)^i t$, $0 \leq i \leq \ell$. Condition (N1) holds because the two first labels are t and $-t$, condition (N3) because the labels come in opposite pairs, and (N4) because the 2^i are distinct. Two ports labelled t and $-t$ satisfy the four conditions directly. For the remaining blocks, and for the assertion that $\{1, 3\}$ admits no neutral labelling, the four conditions are a finite integer feasibility question once a bound on the c_i is fixed, and we settled it by an exhaustive search over integer labels with coefficients at most 8 in absolute value; the resulting labels are listed in Table 1, and each is re-verified from its labels alone, with no reference to how it was found. $\square$

That $\{1, 3\}$ is the one exception does not appear to be an accident of the search. It stays infeasible when the block is allowed three parameters instead of one and coefficients up to 14 in absolute value. We have not proved that no labelling whatsoever works, and we do not claim it; the pair is in any case the one our construction singles out everywhere else as well. A $\{1, 3\}$ block can still be carried by a cell, as the catalogue shows; what fails is only the uniform statement that it may be adjoined to an arbitrary family.

Lemma 3.6 is what keeps the catalogue within reach. A context with a retained neutral unit, or with a pair of recruited ports, is almost always a narrower context with a block adjoined, and the wide contexts are exactly the ones a solver finds hard. Of the 135 contexts our searches had not reached, 128 are obtained in this way from a narrower context that already had a family, at no search cost at all, a further 20 input-in-token variants are obtained the same way, since by Lemma 3.6 the adjunction does not move the pinned input out of the token, and the last 8 contexts, all of them with an $\{o, e\}$ residual and no port, are obtained from a token block. Together with the searches this leaves no context of Lemma 3.8 without a family. The construction is not trusted: the families it produces are written in the format the solver emits and are re-verified by the independent checker of Section 8 together with everything else.

Lemma 3.8 (context completeness). *Every owner receives exactly one context from a finite list, and every vertex of degree two lies in exactly one arm, head chain or neutral unit. Exactly 2,790 of the contexts of the list occur.*

The case analysis is given in Appendix A. Its regression, our executable scheduler, produces a context in the list for every owner of every tree with at most nine vertices (4,782,969 labelled trees) and of 169,000 uniform random labelled trees, 20,000 at each odd order $n \in \{11, 13, 15, 17, 19, 21, 31, 51\}$ and 3,000 at each of $n \in \{101, 201, 401\}$. No owner of any of them

is left without a context and no context outside the list is produced. That the root is a branch vertex is used, and not only for convenience. A vertex of degree two taken as the root can be left with a residual that its arms and its one port cannot cover and with no active child to spend on it, which is the configuration $(h, \{1\}, 1, \text{none})$ at the root; a branch vertex has a third core child and cannot be in that position. On 2,000 random trees of orders 11 to 51, rooting at a vertex of degree two left an owner without a context for 196 of them. Rooting at a branch vertex never did: over 3,000 random trees of orders 11 to 51 we ran the decomposition from *every* branch vertex in turn, 23,000 roots in all, and every one of them gave every owner a context. The choice among branch vertices is therefore free, as Lemma 3.8 states it, and not something the decomposition has to search over. In addition, the decision at a vertex depends only on its head-chain length, its arm lengths, its number of ports and the types of its active children. We therefore enumerated all 5,398,800 abstract local states with up to three arms of lengths at most eight (and larger arm sets of short arms), up to three ports, up to four active children of each of the possible types and head chains of length at most six. Each state was realised by a concrete tree fragment, and we recorded the context emitted at the vertex. The 2,790 distinct contexts obtained are all in the list, and they are exactly the contexts for which families are required in Section 4, whereas the remaining entries of the list are slack.

The cap on the number of active children is the one hypothesis of this enumeration that the decomposition does not force, since an owner of a large tree may carry arbitrarily many active children, and a tree on at most nine vertices carries none with three. We therefore ran the enumeration separately at the caps two, three and four. Raising the cap from two to three adds six contexts, all of them roots with a single arm, and 416 realizable ordered pairs, whereas raising it from three to four adds nothing at all: the 3,818,400 states enumerated at the cap four emit exactly the same contexts and exactly the same ordered pairs as the 1,204,800 states at the cap three. Thus the emitted set is stationary between three and four, which is what the analysis of Appendix A predicts, since the root rule and the two-input rule reduce any number of active children to one or two before a context is emitted. We note that a stationary pair of caps is evidence for, and not a proof of, stationarity at every cap; the proof is the case analysis itself.

Lemma 3.9 (charge). *Let $q \leq D$ be the number of subdivided core edges. The number of owners is at most $2q$, the special support satisfies $|S| \leq 12D + 3$, and the values reserved for forced companions number at most $210q$.*

Proof. Every owner is incident with a subdivided core edge, and such an edge has two end-points. A subdivided edge of length ℓ carries $\ell + 1$ labels, at most $D + q$ in total. Recruited ports contribute at most three per owner, absorptions at most two per owner, and the final bridge triple three. An owner reserves one value for each ratio in $\Phi(F)$ of the family it is given, so the number reserved is at most 10 per owner and at most $210q$ in total, where 10 is the largest number of forced ratios of any family in the catalogue (a choice rule that preferred families with few ratios would lower it to 6, the largest over the contexts of the smallest number in the menu, but our constructor does not use such a rule and we do not rely on one). $\square$

4 Local families

4.1 Symbolic families

A *family* for a context is a list of labels given as integer linear forms, divided by a common denominator L , in parameters y (the head), x or x_1, x_2 (inputs), token coordinates and internal parameters, such that

(E1) $P=O$ holds identically in the parameters;

- (E2) the labels other than the head split into pairs of opposite forms and at most two *tokens*, triples of forms summing to zero (a second token is needed only for two-input cells whose non-head label count is even);
- (E3) the label forms are pairwise distinct and nonzero;
- (E4) the three coefficient vectors of each token span a plane, and a *carrier* of at most two parameters on which they already span a plane is recorded;
- (E5) the head is an independent parameter (nonroot contexts);
- (E6) the only label depending on an input alone is the antipode of that input, and no non-head label is a rational multiple of the head.

We call a family *forced-antipode* when some non-head label is identically minus the head. Such a label necessarily lies in a token $\{-x, a, x - a\}$ of the child, since its mirror is the head and the head is excluded from the signature, and a parent can then absorb the head x neither by a helper port nor by a mirror sibling, because the antipode of the head is already used inside the child. Two contexts of our catalogue carry only forced-antipode families, the (1; one port) bottom and the two-input cell with head chain of length two and residual $\{1\}$, and only the second of them occurs, the first being folded into its parent. Every other context has a family that is not forced-antipode, so a single child context is responsible for the whole of what follows.

A parent absorbs the head of a forced-antipode child through a token of its own that contains the shared edge, $\{x, b, -x - b\}$, so that the two tokens are negatives of each other once the child's carrier is matched, $a = -b$. A stationary parent with at least two ports does this with its port block. An owner uses an *input-in-token* family of its context, found by the same search with the shared edge required to be a token member (and with a second, free token when the number of non-head labels is even). A parent whose context carries no such family instead absorbs the child by joining its rows into its own cell, the joined child type A2 of Appendix A, and the joined cell then has a family that is not forced-antipode, so that it is an ordinary child of the next parent and the absorption stops after one step. Although the two rules look interchangeable, the second is the one that is always available, and the first is what keeps the cells small.

One parent context deserves a separate remark, since it shows that the joining rule cannot be dispensed with. Let v be an owner with head chain of length $\equiv 0$, one arm of length 1, no ports and two kept inputs x_1, x_2 , write a_1, a_2 for its arm labels and h for its head, and read off the three outputs $h + a_1 + x_1 + x_2$, $a_1 + a_2$ and a_2 . The leaf output is a_2 , and $a_1 + a_2 = a_1$ would force $a_2 = 0$, so $a_1 + a_2 = h$ and therefore $h = -(x_1 + x_2)$, which leaves $a_1 = t$, $a_2 = -(x_1 + x_2) - t$ with t free. The four non-head labels admit no token, and pairing t with $-(x_1 + x_2) - t$, or x_1 with x_2 , forces $h = 0$. Thus $t = -x_1$ or $t = -x_2$, and in either case the labels are $-x_1, -x_2, x_1, x_2$. Thus this owner always carries the antipodes of both of its inputs, it has no input-in-token family, and a forced-antipode child under it must be joined. In the joined cell the second child is pinned by one label whose output is that label itself, which is exactly the role of a port, so the joint is searched as the context with one more port and one active child. Property (E6) is what makes adjacent cells compatible without any case analysis, since a parent forces at most the antipode of its input and a child never carries the antipode of its own head, so that no two forms of adjacent cells coincide identically. Property (E6) has two clauses and both are needed for that argument, the first for the parent and the second for the child. We test them together and literally, over every family of every menu and, for a two-input row, against both of its inputs; a separate program does this, since our checker verifies (E1)–(E5) only. By that test, for 141 of the 2,789 contexts that occur and carry a family our catalogue has no family with (E6), and for these the finitely many adjacent pairs

are checked individually by the gate of Section 8. We note that the gate does not rely on this count, since it enumerates every ordered pair of contexts and not only the pairs that (E6) leaves open.

4.2 The search and the checker

Families are found by our constraint model over the integer coefficients of all label forms: one Boolean per (output, label) pair encodes the permutation, each true Boolean imposes the equality of two coefficient vectors, the signature (mirror pairs, token, head, inputs, zero label) is encoded by further Booleans, and every coefficient vector is required to be nonzero and distinct from the others. The model is solved with CP-SAT [9]. A solution is a symbolic family, not a numerical labelling, and it is accepted only after an independent program reconstructs the topology from the row specification, recomputes the outputs, and re-verifies (E1)–(E5) with exact rational arithmetic and at three random integer points. Every family in the table passes, and in addition the six-edge mother and the twelve-label insertion are applied once to every arm and every head chain of every family and the result is re-verified. Table 2 summarises the catalogue.

Four exact impossibilities shape the alphabet. A single arm of length four with no ports has only one-parameter families (rays), so a residual even singleton retains one neutral unit. A head chain of length one or two with ports only has no $P=O$ family at all, and a subdivided parent edge therefore does not by itself make a vertex an owner. The residue-one letter on a direct parent edge has the unique family $(-x; -x-t, t; x)$, whose head is the antipode of its input and is therefore not a free parameter, and it is for this reason joined with its parent. With a subdivided parent edge the same letter has ordinary families with a free head, e.g. $(y; -y-x, -x, y+x; x)$ for $h=1$, and it is then an owner of its own. Finally, the two-input cell with head chain of length two and residual $\{1\}$ has no family without the antipode of its own head, over all denominators $L \leq 12$, all numbers of internal parameters up to three, all coefficient bounds up to 24 and all four token shapes, and this is what makes it the one forced-antipode context that occurs. The classification of the direct-edge two-input row given above is of a different kind, since it is proved by hand and does not depend on the search at all.

5 Sequential realization

Lemma 5.1 (realization). *Let M be the least common multiple of the catalogue denominators and of 24. There is an absolute constant A such that, for every decomposed core with a family assigned to every owner as in Section 4 and $n \geq 10^6(D+1)$, parameter values in $M\mathbb{Z} \cap [-A(D+1), A(D+1)]$ can be chosen bottom-up, and the negatives of the tokens can be placed at stationary vertices of the ordinary part of the tree, so that all special labels are distinct and nonzero, $|S| \leq 30D+3$, and the special support is closed under negation up to at most one zero-sum triple whose negatives are unused.*

Our proof chooses every parameter of every cell numerically, one coordinate at a time, in post-order. The tokens, the zero-sum triples that (E2) leaves after the mirror pairs, are not cancelled against one another: the negatives of each token are placed instead at a *sink* in the ordinary part of the tree, either directly in a zero-sum block of size three or at least five, or, two tokens at a time, in a five-vertex *absorber* built from stationary vertices with two children (Lemma B.1). The hypothesis $n \geq 10^6(D+1)$ is what guarantees enough sinks (Lemma B.4), and a finite check over the catalogue shows that the four extra labels of an absorber never coincide identically with a label of either of its two source cells (Lemma B.5). Thus no symbolic parameter is ever live, apart from the forced-antipode children and the two-input rows that

have their own treatment, the number of forbidden conditions at every choice is $O(|S|)$, and Lemma 3.9 gives boxes of side $O(D+1)$. The full argument is given in Appendix B, together with the explicit value $A = 1.9 \cdot 10^8$, which follows from the catalogue constants (width at most 29, at most 7 parameters, coefficients at most 64 over denominators dividing 12) and Lemma 3.9; nothing is ever divided, so no lattice or coefficient enlargement enters. We have made no attempt to optimise it. An earlier version cancelled the tokens in pairs along the root path; why this cannot give a linear threshold is recorded in Remark B.6.

6 Dense completion and the proofs of Theorems 1.1 and 1.2

After Lemma 5.1 the unused labels form the set $X = \mathbb{Z}_n \setminus (\{0\} \cup S)$, which is closed under negation up to at most one leftover triple T (the case of an odd number of tokens). The negatives $-T$ are placed in an odd ordinary block of size at least three, which exists by parity whenever such a triple is left over and no block has size one; write r for that block's prescribed size and put $X' = X \setminus (-T)$. Since r is odd and at least three, the remainder $r-3$ is zero or at least two, and the block is dropped from the list when it is zero. It remains to partition X' into zero-sum blocks of the resulting sizes, all at least two, with $|\mathbb{Z}_n \setminus X'| \leq 2A(D+1) + 4$.

Lemma 6.1 (linear dense completion). *There are absolute constants $\eta > 0$ and n_1 such that for every odd $n \geq n_1$, every symmetric $X \subseteq \mathbb{Z}_n \setminus \{0\}$ with $|\mathbb{Z}_n \setminus X| \leq \eta n$, and every list of block sizes $r_w \geq 2$ summing to $|X|$, the set X has a partition into zero-sum blocks of sizes r_w .*

Proof of Lemma 6.1. Write $n = 2K + 1$. Since n is odd and X is symmetric with $0 \notin X$, the set X is a disjoint union of inverse pairs $\{j, -j\}$; let $J \subseteq [1, K]$ be the set of magnitudes occurring, so that $|X| = 2|J|$ and $|\mathbb{Z}_n \setminus X| = 1 + 2(K - |J|)$. In particular $|X|$ is even, so the number o of odd block sizes is even. Let m_2 be the number of blocks of size two and put $N = \sum_{r_w \geq 3} r_w$, so that $|X| = N + 2m_2$. We fix $\eta \leq 1/20$ and argue by cases according to the size of N .

Case 1: $N \leq n/100$. In this case we use no external input. A *twin* is a pair of magnitudes $a < b$ in J with $a+b \in J$; it yields the two zero-sum triples $\{a, b, -(a+b)\}$ and $\{-a, -b, a+b\}$, which are disjoint and whose union $\pm\{a, b, a+b\}$ is symmetric. We choose $o/2$ twins with pairwise disjoint magnitude sets, greedily. At the start of step s the forbidden magnitudes are those outside J , of which there are at most $\eta n/2$, together with the $3(s-1) \leq 3o/2 \leq N/2 \leq n/200$ already used. Hence at most $(\eta + 1/100)K + 2$ magnitudes are forbidden. On the other hand the number of triples $\{a, b, a+b\} \subseteq [1, K]$ with $a < b$ is $\sum_{c=3}^K \lfloor (c-1)/2 \rfloor \geq K^2/5$ for K large, whereas a single magnitude lies in at most $2K + K/2 + 3$ of them. Since $(\eta + 1/100) \cdot (5/2) < 1/5$ for $\eta \leq 1/20$, a permissible twin survives at every step once n is large, although the margin is not optimised. Now pair the odd blocks arbitrarily, give the two triples of the s -th twin to the two blocks of the s -th pair, and distribute the remaining magnitudes as inverse pairs, $p(r_w) = \lfloor r_w/2 \rfloor - [r_w \text{ odd}]$ of them to the block w . The count is exact, because $\sum_w p(r_w) = (|X| - 3o)/2 = |J| - 3o/2$ is precisely the number of magnitudes left over. Every block then has its prescribed size and sum zero.

Case 2: $N > n/100$. Here we run the argument of [7, Lemma 6.27] in $\mathbb{Z}_n$, which has no involutions, and with the missing pairs held out. Put $p = N/(2|J|)$, so $1/100 \leq p \leq 1$ because $2|J| = |X| \leq n$. Enumerate all K inverse pairs of $\mathbb{Z}_n$ and let Y be a p -random set of indices; by [7, Lemma 6.26] applied to the partition of $\mathbb{Z}_n$ into these pairs together with $\{0\}$, the union of the selected pairs splits as $Q \sqcup R \sqcup S$ with Q, R disjoint $p/2$ -random subsets of $\mathbb{Z}_n$ and $S \subseteq \{0\}$. Let Y_J be the set of selected indices whose magnitude lies in J ; it is p -random on $|J|$ indices with integral mean $p|J| = |J| - m_2$, so its median equals its mean and, with probability at least one half, $|Y_J| \leq |J| - m_2$, while Chernoff's bound gives $|Y_J| \geq |J| - m_2 - \sqrt{n} \log n$ with high probability. Fix an outcome in which both hold and in which Q and R satisfy the conclusions

of [7, Lemmas 6.23 and 6.25]. Reserve m_2 of the $|J| - |Y_J|$ unselected available pairs for the blocks of size two, let J^* be the union of the at most $\sqrt{n} \log n$ remaining unselected available pairs, and put $Z = J^* \cup \bigcup_{i \in Y_J} \{i, -i\}$, a union of inverse pairs with $|Z| = 2|J| - 2m_2 = N$ and $\sum Z = 0$.

Reduce the multiset $\{r_w : r_w \geq 3\}$ to a multiset $M' \subseteq \{3, 4, 5\}$ as in the first paragraph of the proof of [7, Lemma 6.25], recombining at the end, and split $M' = M^Q \sqcup M^R$ by giving M^Q the floor and M^R the ceiling of half the multiplicity of each of 3, 4, 5. The multiplicities need not be even, so the two parts need not have equal sums; however they differ in multiplicity by at most one in each of the three sizes, hence $|\sum M^Q - \sum M^R| \leq 12$, and a bounded discrepancy is absorbed by the slack reserved below.

Write $\delta = (p/2)^{10^{10}} / \log(n)^{10^{24}}$ for the perturbation radius that [7, Lemma 6.25] allows at density $p/2$, and let $\varepsilon = \delta/24$, which lies in the range permitted by [7, Lemma 6.23] once n is large. Both $Q \setminus Z$ and $R \setminus Z$ are contained in $\{0\} \cup (\mathbb{Z}_n \setminus X)$ and hence have size at most $|\mathbb{Z}_n \setminus X|$, and $|\sum M^Q - pn/2| \leq |\mathbb{Z}_n \setminus X| + O(\sqrt{n} \log n)$. Assume $|\mathbb{Z}_n \setminus X| \leq \delta n/48$, which is the hypothesis of the lemma with $\eta = \delta/48$. The two hypotheses of [7, Lemma 6.23] are then met and not merely its conclusion is used: $|Q \setminus Z| \leq |\mathbb{Z}_n \setminus X| \leq \varepsilon n/2$, and $|\sum M^Q - pn/2| \leq \varepsilon n/2 + O(\sqrt{n} \log n) \leq \varepsilon n$ once n is large. Then [7, Lemma 6.23], applied with $R = Q$, Z as above, $m = \sum M^Q$ and $g = 0$, returns $Q' \subseteq Z$ with $|Q'| = \sum M^Q$, $\sum Q' = 0$ and $|Q' \triangle Q| \leq 6\varepsilon n = \delta n/4$. Put $R' = Z \setminus Q'$; then $|R'| = \sum M^R$, and $\sum R' = \sum Z - \sum Q' = 0$, and

$$|R' \triangle R| \leq 6\varepsilon n + |\mathbb{Z}_n \setminus X| + |J^*| \leq \frac{\delta n}{4} + \frac{\delta n}{48} + O(\sqrt{n} \log n) \leq \delta n$$

once n is large, and likewise $|Q' \triangle Q| \leq \delta n/4 \leq \delta n$. Thus both perturbations stay strictly inside the allowance, which is the point of taking ε a twenty-fourth of δ rather than a sixth, while the deletion radius $\eta = \delta/48$ is what keeps the two *input* hypotheses inside εn ; the two budgets are separate and an earlier version of this proof conflated them. Therefore [7, Lemma 6.25] applies to Q' with the reference set Q and the multiset M^Q , and to R' with R and M^R , and returns zero-sum partitions of Q' and of R' of the prescribed sizes. Together with the m_2 reserved inverse pairs these fill every block, since $Z \sqcup (\text{reserved pairs}) = X$. When $m_2 = 0$ one has $p = 1$, Y_J is all of J and J^* is empty, and the argument degenerates to a direct application of [7, Lemma 6.25] to X . Thus X has the required partition whenever $|\mathbb{Z}_n \setminus X| \leq \delta n/48$, and δ depends only on $p \geq 1/100$ and on n through the logarithm. $\square$

Remark 6.2 (the two deletion bounds of [7]). Two readings of [7] enter through the parameter ε of the last proof, and they give the two thresholds of Section 1. The printed perturbation radius of their Theorems 4.3–4.7, and hence of their Lemmas 6.23 and 6.25, is $p^A n / (\log n)^A$; since the proof above uses those lemmas only with $p \geq 1/200$, this gives Lemma 6.1 with ηn replaced by $n / (\log n)^A$, and hence Theorem 1.1. Their Section 7, under *Bounds in the main theorem*, records that $\log n$ may be replaced by $\log |G'|$ throughout the proof, the only source of logarithms being their short commutator-product theorem, and for an abelian group the commutator subgroup is trivial and the commutator product is empty; the authors record the consequence $g(n) = \Omega(n)$ for cyclic groups in their Problem 7.4. One should read their commutator length as bounded by $\max(1, \log_4 |G'|)$ rather than divide by $\log 1$. Their Lemmas 6.23 and 6.25 are deduced from Theorems 4.4–4.6 with no further logarithmic loss, the auxiliary error terms being of order $\sqrt{n} \log n$, so at every fixed reservoir density this reading leaves a positive constant perturbation radius and gives Lemma 6.1 as stated, which is what Theorem 1.2 uses. The authors describe this improvement as requiring essentially no modification of their proof and state its consequence for cyclic groups, $g(n) = \Omega(n)$, in their Problem 7.4; neither is a numbered theorem in their paper, and we have asked the authors to confirm the linear form of Lemmas 6.23 and 6.25. The constant η is in either case inherited from their argument and is not effective.

Proof of Theorems 1.1 and 1.2. Root T at a branch vertex, decompose by Section 3, assign families by Section 4, realize by Lemma 5.1 with $|S| \leq 30D+3$ and magnitudes at most $A(D+1)$ (the hypothesis $n \geq 10^6(D+1)$ of that lemma is implied by $n > C(D+1)$ once $C \geq 10^6$), and place the leftover triple if there is one. With the linear deletion bound of Remark 6.2, Lemma 6.1 completes the labelling as soon as $2A(D+1)+1 \leq \eta n$; every stationary vertex then has a zero-sum block, every special vertex has the prescribed sum, and Lemma 2.1 gives the bijection, so $C \geq 3A/\eta$ and $n_0 \geq n_1$ prove Theorem 1.2. With the printed perturbation radius of [7] in place of the linear one the same argument applies when $2A(D+1)+1 \leq n/(\log n)^{A'}$, which proves Theorem 1.1; its final assertion follows by taking $D \leq k$ and n large. $\square$

7 An elementary route to the linear threshold

Theorem 1.2 rests on the linear deletion bound for cyclic groups of [7] through Lemma 6.1 (Remark 6.2). Here we reduce Lemma 6.1 to a single statement about permutations of an interval, prove that statement at the two ends of its range and for a family of composite orders, and verify it exactly up to order 30. A proof in general would give Theorem 1.2 unconditionally, with an explicit C .

7.1 From the residual set to Heffter's difference problem

After Lemma 5.1 the special support is $\pm I$ with $I \subseteq [1, M]$ and $M = A(D+1)$, and the residual set is $X = \pm([1, K] \setminus I)$ in $\mathbb{Z}_n$, $n = 2K+1$. In the branch of Lemma 6.1 that needs the completion theorem, X has to be partitioned into signed zero-sum triples $\{\pm a, \pm b, \pm c\}$. Writing a triple by its three magnitudes $a < b < c$ in $[1, K]$, the two possibilities are $a+b=c$ and $a+b+c=n$; the first is a pair $\{b, c\}$ at distance a and the second is the pair $\{b, -c\}$ at cyclic distance a in $\mathbb{Z}_n$. For $I = \emptyset$ this is Heffter's first difference problem when $n \equiv 1 \pmod{6}$, solved for every n by Peltesohn [8], and the general case asks for the same partition of $[1, K] \setminus I$, that is, for Heffter's problem with the small differences I forbidden. The hardest case in our experiments is $I = [1, M]$, and it is the one we analyse.

7.2 Folded sum permutations

Let $K-M=3t$. In every solution our searches found, the t smallest elements $M+1, \dots, M+t$ can be taken as the distances a , and the remaining $2t$ elements split into a lower half $M+t+1, \dots, M+2t$ and an upper half $M+2t+1, \dots, K$ with each pair $\{b, c\}$ taking b from the lower and c from the upper half. Writing $a = M+1+i$, $b = M+t+1+j$, $c = M+2t+1+k$ with $i, j, k \in [0, t-1]$ and $s = M+1-t$, the two triple types become

$$c-b=a \iff k=i+j+s, \quad b+c=n-a \iff k=(2t-1)-(i+j+s).$$

So a solution is a permutation π of $[0, t-1]$, $j = \pi(i)$, such that the numbers $u_i = i + \pi(i) + s$ lie in $[0, 2t-1]$ and their folds $\min(u_i, 2t-1-u_i)$ form a permutation of $[0, t-1]$. We call such a π a *folded sum permutation* of order t and shift s , and write $\text{FSP}(t, s)$. Equivalently, with $\tau_i = i + \pi(i)$, the τ_i are distinct, lie in the window $W = [-s, 2t-1-s]$, and no two of them sum to the reflection constant $c = 2t-1-2s$ of W ; that is, the sum set of π is a transversal of the reflection of W . Since $\sum_i \tau_i = t(t-1)$ for every permutation, the mean of the transversal exceeds the centre of W by $s - \frac{1}{2}$.

The range $3M+1 \leq K \leq 7M+3$ of Lemma 6.1 corresponds to $(1-t)/2 \leq s \leq (t+1)/2$, and beyond $K \geq 7M+3$ Simpson's theorem [10] on Langford sequences applies with no folding at all.

Conjecture 7.1. *For every $t \geq 5$ and every integer s with $(1-t)/2 \leq s \leq (t+1)/2$ there is a folded sum permutation of order t and shift s .*

The window is sharp: outside it no $\text{FSP}(t, s)$ exists, because the s largest values of W (for $s \geq 1$) have no reflection partner in the range of the τ_i and must all be attained, which is impossible once $s > (t+1)/2$. Our constraint programs verify the conjecture exactly, for every $3 \leq t \leq 30$ except $t = 4$, where $s = -1$ and $s = 2$ fail; unlike Simpson's theorem there are no congruence exceptions inside the window, the folding removes them.

7.3 What is proved

Proposition 7.2. *Conjecture 7.1 holds in the following cases.*

1. $s \in \{0, 1\}$: the identity is an $\text{FSP}(t, s)$.
2. t odd and $s \in \{(1-t)/2, (t+1)/2\}$: the rotation $\pi(i) = i + (t-1)/2 \pmod t$ is an $\text{FSP}(t, s)$ at both endpoints.
3. (Composition.) If a is odd and σ is an $\text{FSP}(b, s')$, then

$$\pi(au + w) = a\sigma(u) + ((w + \frac{a-1}{2}) \pmod a), \quad u \in [0, b-1], \quad w \in [0, a-1],$$

is an $\text{FSP}(ab, s)$ with $s = a(s' - 1) + (a+1)/2$.

Proof. (1) The sums $2i$ are distinct and even, the constant c is odd, and $0 \leq 2i \leq 2t - 2 \leq 2t - 1 - s$ exactly when $s \leq 1$; the lower bound $2i \geq -s$ needs $s \geq 0$.

(2) For the rotation the sums $i + \pi(i)$ are $2i + (t-1)/2$ for $i < (t+1)/2$ and $2i - (t+1)/2$ for $i \geq (t+1)/2$, which together form the interval $[(t-1)/2, (3t-3)/2]$. At $s = (1-t)/2$ this interval is the lower half of W , at $s = (t+1)/2$ it is the upper half, and a half of W contains no reflected pair.

(3) On each block the inner rotation has sums $w + \rho(w)$ forming the interval $I = [(a-1)/2, (3a-3)/2]$, which is carried to itself by $x \mapsto 2a - 2 - x$. The sums of π are $a(u + \sigma(u)) + I$. Put $s = (a+1)/2 + aq$. The condition $a(u + \sigma(u)) + I \subseteq [-s, 2ab - 1 - s]$ is $u + \sigma(u) \in [-(q+1), 2b - q - 2]$, the window of $\text{FSP}(b, q+1)$; and the reflection constant $c = 2ab - 1 - 2s = a(2b - 2q - 1) - 2$ sends $ay + x$ to $a(2b - 2q - 3 - y) + (2a - 2 - x)$, so it reflects I onto itself and the block index y by the constant $2b - 1 - 2(q+1)$ of $\text{FSP}(b, q+1)$. Therefore the sum set of π is a transversal exactly when the sum set of σ is, and $s' = q+1$. $\square$

Case (2) has two readings: at the lower endpoint it is a hooked Langford sequence of defect $(t+1)/2$, at the upper endpoint the columns $(i, \pi(i), c - i - \pi(i))$ form a $3 \times t$ magic rectangle. Case (3) covers the residues $s \equiv (a+1)/2 \pmod a$ for every odd divisor a of t ; it does not reach prime t , and we have not found an additive counterpart.

7.4 Consequences

If Conjecture 7.1 holds then the clean tail $[M+1, K]$ has a maximum packing into difference triples for every $K \geq 3M+1$ (the cases $K - M \not\equiv 0 \pmod 3$ reduce to it by omitting the one or two smallest elements, and the finitely many cases $K \leq 3M+3$ are magic rectangles with one or two empty cells, verified for $M \leq 30$). Together with Simpson's theorem on Langford sequences, which covers $K \geq 7M+3$ with no folding, this gives Lemma 6.1 with $\eta = 1/(14A + O(1))$, hence Theorem 1.2 unconditionally, with an explicit C and without any use of [7]. The punctured cases $I \subsetneq [1, M]$ are the same object with the extra small elements as additional distances, and were found feasible for every random I we tried; we have not reduced them to Conjecture 7.1 in general.

8 Computations and certificates

Three programs enter the proof, and we describe them in the order in which their results are used.

The symbolic search. For a context the search builds the label forms of Section 4 with integer coefficients bounded by 24 over a common denominator $L \leq 12$ (the families built by the two identities of Section 3 are not subject to this cap, and the largest coefficient anywhere in the catalogue, 64, is attained by the equal-pair blocks; it is this catalogue-wide value that enters Lemma 5.1), encodes the $P = O$ identity as a permutation of the output multiset onto the label multiset (one Boolean per pair, each true Boolean equating two coefficient vectors), encodes the signature (E2), the distinctness (E3), the token rank condition (E4) and the head freedom (E5) as further Boolean and linear constraints, and asks CP-SAT [9] for a feasible assignment. The same model with the cleanliness conditions (E6) is tried first, and the alternatives (token of rank three, token of rank two, no dedicated token parameters, two tokens) are tried in that order. Every feasible assignment is stored, and the menu of a context lists all of them in order of preference, families containing the antipode of their own head last. For the contexts with inputs the model is run a second time with the shared edge required to be a token member and the antipodes of the inputs forbidden (and a third time, for two-input cells, with both inputs required to be token members). The resulting *variant* families are kept in separate tables and are the ones used when a child is forced-antipode. Each context is tried under nine settings of (denominator, number of internal parameters, coefficient bound), namely (1, 1, 3), (2, 1, 3), (3, 1, 3), (1, 2, 4), (2, 2, 4), (3, 2, 4), (4, 2, 4), (6, 2, 6) and (12, 2, 12), with the head multiplier free, and with a time limit between 150 and 900 seconds per context and setting on a 128-core workstation. In all, 14,047 contexts with widths up to 29 labels were attempted, in 30,485 runs of the solver, and 5,446 of them received at least one family. The search is exact in the sense that an infeasible model is a proof of non-existence within the coefficient and denominator bounds, and it is in this sense that the impossibilities of Section 4 are stated.

The independent checker. The checker reads only the JSON record of a family, rebuilds the cell topology from the row specification (head chain, arms, ports, active and input children, root row), recomputes every output as an exact rational combination of the parameters, and verifies (E1) by matching the two multisets of coefficient vectors, (E2)–(E5) directly, and the numerical identity at three random integer points, and it shares no code with the search. All 12,748 families of the catalogue and all 2,588 variant families pass, as do the families obtained from them by one six-edge insertion on every arm and by one twelve-label insertion on every head chain. The command

```
python3 scripts/check_alphabet_families.py data/alphabet_families.json
```

re-verifies the whole catalogue in about two minutes on a laptop and prints one line per context.

The gate and the closure check. Two adjacent cells can fail to be compatible in only one way. A label of the parent that depends on its input alone is qx for a rational q , a label of the child that depends on its own head alone is $q'y$, and if the two ratios agree once the shared edge is identified then the two labels are equal as forms and no choice of parameters separates them. Our gate enumerates every ordered pair of contexts, tries every pair of families from the two menus, and reports the context pairs for which no combination avoids such a coincidence. Since the sink route places the negative of every unit label elsewhere in the tree, the gate also checks the negative of each unit label of either cell against the labels of the other (this

is what Lemma B.5 needs for the cells that violate the second clause of (E6)); the addition changes the count by one pair, which is not realizable. Over the catalogue it reports 0 ordered pairs. However, most ordered pairs of contexts cannot occur in a tree at all, and restricted to the 11,927 ordered pairs that the enumeration of Lemma 3.8 records it reports 0. All but one of these have the forced-antipode child of Section 4 as their child and are therefore absorbed by the rule of that section and not by the gate. The exception appeared only when the enumeration was run at three active children, and it is the pair whose parent is the root context with a lone arm of residue two and one active child, and whose child is the bottom cell with arms of lengths one and two. Here the parent's forced ratios are -1 , $-\frac{1}{2}$ and $\frac{1}{2}$ in every family of its menu, and the child carries the antipode of its own head in every family of its menu, so the two collide at every head multiplier the search produces. However, the gate is a sufficient condition and not a necessary one: we join the child into the parent and solve the two label sets together as a single cell of width ten, for which the catalogue carries a family, and an identity between labels of one cell cannot arise. We note that the joining stops there, since the parent is the root and the child is a bottom cell, so the enlarged cell has neither a parent of its own nor an active child, and this is the same device as the joint of the absorption rule. A second program checks the conditions that the absorption rule needs, namely that every context that occurs has a family, that every context with a free active child has an input-in-token family or a joint that is not forced-antipode, that the folded letter has such a joint, and that every two-input context either has a family with both inputs in tokens or has a port joint carrying the second pinned child in a token. We note that our scheduler deliberately stops one step short of the absorption rule, since it emits the forced-antipode cell as an owner of its own and does not join it into its parent, because which of the two routes applies depends on the certificate table and not on the tree. The context list therefore carries the joint of every free-child context and the port joint of every two-input context, and the second program is what verifies that the rule closes over them. These conditions are a finite list and the program reports them one by one, so that the state of the verification is visible at any time. All of them hold: every one of the 2,790 contexts that occur has a family, all 544 free-child contexts absorb a forced-antipode child, the folded letter and all 40 two-input contexts do the same, and the gate leaves nothing outside the absorption rule and the single join above.

The scheduler and the constructor. Our scheduler implements the rules of Appendix A and returns, for a rooted tree, the list of owners with their contexts. It is the regression for Lemma 3.8 and was run on all labelled trees with at most nine vertices and on 169,000 random trees of odd orders between 11 and 401. It roots the core at a branch vertex, as Lemma 3.8 requires; an earlier version of it took the root from its caller without checking the degree, which is what produced the only uncovered owners we have seen. Our constructor implements Lemma 5.1 on top of the scheduler and the catalogue, produces an integer labelling, reduces it modulo n , and checks the bijection of vertex sums directly. Each run writes a certificate listing the owners, the chosen families, the parameter values and the final labelling, so that a labelling can be re-verified without the constructor. The constructor has two modes. In its default mode it follows the decomposition but completes the ordinary blocks by a direct search, and in the sparse regime ($n \in \{51, 101, 201\}$ with $D \approx n/25$) it succeeded on 116 of 120 random trees. Its *proof-faithful* mode follows the earlier version of Lemma 5.1, in which tokens were cancelled in pairs along the root path (numeric coordinates, live carriers, deferred pairing, the reservation of forced companions), and reports every departure from that lemma as a named gap. On the same 120 trees it succeeded on 98. Of the twenty-two remaining runs, fourteen report a modular collision between two special labels, or a pending triple whose negatives are already used modulo n , seven report a failure of the completion step, and one reports that no odd block was available for the singleton bridge. On a comb attached to a

dense part, however, the same mode records the gap “a third token became pending” at every second core vertex of the comb, with head labels doubling along it, which is the failure of the pairing route recorded in Remark B.6. A third mode (`sink_route/sink_constructor.py`) implements Appendix B as it now stands: it reserves the sinks before any label is chosen, chooses every parameter numerically, fills the sinks and completes the ordinary blocks. It labels the comb attached to a root with $12k$ stationary children for $k \in \{5, 9, 15\}$, the comb attached to a complete binary tree of depth 7, 8 and 9 for the same three values of k (using 39 absorbers and 9 direct sinks in all), and fifteen random sparse trees of orders 101, 201 and 401, with no lemma gap on any of these 27 trees; the largest label magnitude is below $7(D+1)$ and $|S| < 4.5(D+1)$, against the bounds $A(D+1)$ and $30D+3$ that we prove (with units in place of tokens the same trees give the same outcomes, four of the sparse trees now sending a quad to a block of size four, and the quad gadget, the sink for the second token of a forced-antipode family and the exclusion of a parent family above a residue-two ray were each exercised on a small tree built for the purpose). Whether a tree has a cancelling triple at all depends on the root: none of the trees above has one at the root the constructor chooses first, so the triples are exercised separately. On the sparse tree of order 201 rooted at the vertex which does produce one, the three heads 360, -4320 , 3960 sum to zero, their negatives are placed together in one ordinary block, and the labelling verifies (`sink_route/regression_cancel_triple/`). On the 293 free trees of orders 7, 9 and 11 it succeeds on 243; of the fifty 293 free trees of orders 7, 9 and 11 it succeeds on 243; of the fifty failures, 33 are modular collisions, nine report that the tree has no sink at all (no block of size three or at least five and no path of five stationary vertices with two children), which is the hypothesis of Lemma 5.1 failing, as it must at these orders, and eight fail in a cell or in the completion. Every successful certificate was re-verified independently of the constructor. The same mode realizes the two-level joints of Lemma B.7 (`sink_route/reset_joint_constructor.py`): on the comb of Appendix B whose every core vertex has two active children and one arm of length one, attached to a bushy root with $n \approx 12.5(D+1)$, chains of 4 to 8 rows are cut into 2, 2, 3, 3 and 4 joints and every tree is labelled with the largest magnitude below $6(D+1)$, whereas without the joints the same trees either fail or reach $21(D+1)$; the scheduler’s owner lists on all trees of order at most nine are unchanged by the addition. We note that all these runs use the dense completion only through a direct construction of the leftover blocks (from the antipodal pairs, which the sink route makes available, with the CP-SAT search as a fallback), since at these orders the asymptotic Lemma 6.1 does not apply, and their purpose is to confirm the bookkeeping of the cells and of the sinks and not the theorem itself. A modular collision between two special labels is exactly the event that the hypothesis $n > C(D+1)$ excludes, and at $n = 201$ with $D \approx 8$ the hypothesis is very far from holding.

The absorber gate and the sinks. Lemma B.5 needs, for every pair of cells that can share a parameter, a role order of the two tokens for which none of the four absorber forms coincides identically with a label of either cell, with the negative of one, with another absorber form or with zero. Our program `sink_route/absorber_gate.py` checks this over the joint parameter space of the two families, the child’s head identified with the parent’s input, for every ordered pair of a catalogue family with one input and a token (625 families) and a catalogue family with a head and a token (2,113 families), one family per context, namely the one the scheduler prefers there: of the 1,320,625 pairs none has fewer than eight collision-free orders among the thirty-six, and 1,077,257 have all thirty-six. The same check over the 4,242,904 pairs with a two-input parent (the child attached at either input, either token of the parent), the 574,907 pairs in which at least one side uses its second token, and the 401 families carrying two tokens (both tokens of one cell as the two sources) finds no pair without a collision-free order either, the minima being six, eight and three orders respectively (`sink_route/absorber_gate_ext.py`). The same test was then run over *every* entry of every

menu rather than one family per context (`sink_route/full_menu_absorber_gate.py`): 2,175 parent families against 4,431 child families, 9,637,425 ordered pairs, again no pair without a collision-free order, the minimum being six. A second implementation, which carries the labels as exact rationals over an explicit joint coordinate list and shares no code with the first, reproduces that run pair for pair, including the whole histogram of collision-free orders (`sink_route/verify_full_menu_gate.py`); the certificate records the catalogue’s SHA-256 digest. Thus the pairing of Lemma B.5 may be arbitrary. The units themselves were scanned over the 6,975 token-bearing families of all menus: 528 contain an antipodal pair among their token labels, of which 394 leave a quad and 134 leave nothing to place, and every family with $-x$ in a token also has x in a token, so the case the units were introduced for never arises in any other form. For the residue-two ray, exactly one single-input family (`h4_6-6_p0_act`, first of its menu) has a label $\pm x/3$ or $\pm 2x/3$ and its context has another family without, and one two-input joint family (`h3_1-3-3-3_p0_A2`) has such a label on one of its two inputs, to which the constructor never assigns a ray; no context is left without an admissible family. The counting of Lemma B.4 was also run on real schedules (`sink_route/packing_statistics.py`): on random sparse trees with $D \approx n/25$ the direct capacity alone exceeds the number of owners by a factor of at least twelve, on a comb of k core vertices attached to a root with $12k$ stationary children the root’s block absorbs everything, and on a comb attached to a complete binary tree of depth d , where no block has size three or more, the greedy packing finds 2^{d-4} disjoint absorbers, enough for every $k \leq 15$ from $d = 7$ on. These runs are evidence for the constants, not part of the proof.

9 Discussion

Previous work on edge-graceful trees is of two kinds. Explicit labellings are known for several families, for example paths, stars, spiders and brooms (see [2] for the list; the regular spiders of [1] already carry unboundedly many vertices of degree two, although in a single prescribed shape), and the zero-sum partition theorem of [3] covers every odd-order tree with at most one vertex of degree two. Our earlier work [6] extended that theorem to two such vertices with explicit (and effective) constructions. Our result lies between the two: it covers all trees in which the vertices of degree two are a small fraction of the order, with no restriction on where they sit, at the price of an ineffective constant.

The constant C is not effective because Lemma 6.1 is not, whereas the constant A of Lemma 5.1 is explicit ($A = 1.9 \cdot 10^8$). An effective version of the zero-sum matching with linear deficiency would therefore make Theorem 1.2 effective without any change to the construction, and Section 7 reduces this to Conjecture 7.1. However, the degree-two vertices remain a bottleneck in a stronger sense: the special support has size linear in D because every degree-two vertex is non-stationary, so our argument cannot reach trees in which these vertices have positive density. In particular it says nothing about caterpillars with many subdivided edges or about spiders with long legs, for which the explicit constructions in the literature remain the only tool. Whether the density threshold can be raised, e.g. to $D = o(n)$ or to $D \leq cn$ for a small absolute c , seems to us the natural next question. The local families are indifferent to the density, so the obstacle is entirely in the completion step, where the leftover set would no longer be nearly the whole group.

Two further remarks concern the computer-assisted part. First, our catalogue is finite by design and every family in it is a rational identity in a few parameters, and an independent re-verification is for this reason possible at all. The search is only a way of finding the identities, and the proof does not depend on the search having been exhaustive except in the four impossibility statements of Section 4, each of them a finite enumeration, and one of the four structural facts we use, the classification of the direct-edge two-input row, is proved by

hand. Second, although our scheduler’s regression covers all small trees, it is evidence for, not a proof of, Lemma 3.8. The proof is the case analysis of Appendix A, and the scheduler is our way of keeping that analysis honest.

10 AI-assisted research statement

Anthropic’s Claude, OpenAI’s ChatGPT, Google’s Gemini and Moonshot AI’s Kimi were used as coding and analysis assistants throughout this project. They implemented and executed the processing, reasoning, and statistical code, prepared documentation, surveyed literature, and assisted in preparing manuscript text. The author assumes responsibility for all content.

11 Data and code availability

The search (`symbolic_free_token_cpsat.py`), the batch driver, the independent checker (`check_alphabet_families.py`), the scheduler (`context_scheduler.py`), the abstract enumeration (`enumerate_emittable_contexts.py`), the gate (`check_lookahead_gate.py`), the closure check (`closure_check.py`), the literal test of (E4) and (E6) (`audit_e4_e6.py`, `e6_coverage.py`), the non-constancy check of the absorber gate and the sink statistics (`sink_route/absorber_gate.py`, `sink_route/full_menu_absorber_gate.py` with its independent verifier, `sink_route/packing_statistics.py`), the constructors (including `sink_route/sink_constructor.py` with its regression driver), the context list, the certificate catalogue (`alphabet_families.json`) with its variant tables, and the certificates of the runs reported in Section 8 are available at <https://github.com/lsmeng/sparse-edge-graceful> and archived at <https://doi.org/10.5281/zenodo.22287387> (concept DOI, resolving to the latest version).

A Context completeness

We prove Lemma 3.8 by describing the assignment step by step. The executable scheduler follows the same steps and is our regression for the case analysis. Process the core bottom-up and let v be a nonroot core vertex with parent core edge of length h , arm multiset A , k active children (children which are owners) and q ports. Since v has degree at least three in T we have $|A| + k + q \geq 2$.

Step 1 (arms). Lemma 3.1 partitions A into neutral units and a residual R of one of the six types. The units are $P=O$ with zero head sum and with support closed under negation, so they do not enter the cell of v . However, when a unit is needed at v itself (Steps 3 and 4) one of them is retained and its arms are added to the residual.

Step 2 (active children). Cancelling a set of $j \geq 2$ active children means prescribing their heads to sum to zero: for j even we use $j/2$ mirror pairs, and for j odd one free zero-sum triple together with $(j-3)/2$ pairs. These heads are free parameters of the children’s families by (E5), with one exception, and a child of the joined types, whose head is forced, is paired with a sibling of free head, the sibling taking the negative of the forced head. The exception is a pair of two children that both have forced heads, which our scheduler does produce. It is also arrangeable, one level further down: a child whose head is forced in this way has context $(0, \{1\}, 0, \text{free})$, its own active child is of free type, and the head of that grandchild is a free parameter by (E5), so setting the grandchild’s head makes the two forced heads negatives of one another. We note that this uses (E5) at the grandchild and not at the child, and that it

does not recurse further, since the grandchild is of free type and is therefore not one of the rigid rows.

Whether all of the active children can be cancelled depends on what Steps 3 and 4 can then do with the residual. If cancelling all of them leaves v covered, we cancel all of them, and v has at most one active child left, whose head is the input x . Otherwise v must spend active children on the residual, and the rule is by the parity of their number k . For k even we keep two and cancel $k - 2$, so that v is a *two-input* owner with inputs x_1, x_2 ; for k odd we keep one and cancel $k - 1$. In both cases the number cancelled is at least two whenever it is positive, so the cancellation above applies. At the root with a lone residual arm there is one deviation: $k = 2$ keeps both, because keeping one would leave a single child to cancel, and every $k \geq 3$ keeps one.

The configurations in which the residual cannot be covered without active children, so that this rule is used, are the following, and they are the exact condition our scheduler tests. Off the root, v has no ports, its residual is a single arm, and no neutral unit was retained; then the residues modulo six that need an active child are 1, 2, 4, 6 when the head chain has length 0 modulo six and all of 1, $\dots$, 6 otherwise. At the root with a lone residual arm of residue r and q ports, the recruited port count of Step 5 has no feasible value exactly when $q = 0$ and $r \in \{2, 4\}$, or $q = 1$ and $r \in \{1, 3, 5\}$; for $q \geq 2$ there is always one. At the root there is one further case, the opposite-parity residual $\{1, 2\}$ with no ports. We note that an earlier version of this step named only the portless residuals $\{1\}$ and $\{2\}$ and only $k = 2$; that was too narrow, as a tree of order 19 on which our scheduler keeps two active children at a residual of residue four shows. With a subdivided parent edge such cells have ordinary families, e.g. $(y, -y - x_1 - x_2; -x_1 - x_2, y + x_1 + x_2; x_1, x_2)$ for $h = 1$ and $R = \{1\}$ with token $\{x_1, x_2, -x_1 - x_2\}$, some of them with two tokens. With a direct parent edge the exact search shows that no family with a free head exists, and v uses instead one of the two *rigid* rows $(-x_1 - x_2; -x_2, -x_1; x_1, x_2)$ for $R = \{1\}$ and $(-\frac{3}{2}(x_1 + x_2); -\frac{1}{2}(x_1 + x_2), -(x_1 + x_2), \frac{1}{2}(x_1 + x_2); x_1, x_2)$ for $R = \{2\}$, whose head is forced by the inputs and which are treated like the residue-two ray in Appendix B. In our random-tree regression this pattern occurs at roughly one to two percent of the vertices.

Step 3 (empty residual). Suppose $R = \emptyset$. If no active child remains, v is stationary and its ordinary block consists of its q ports, which requires $q \neq 1$. If $q = 1$ there is a neutral unit or a cancelled group because $|A| + k + q \geq 2$, and one neutral pair is undone, so that R becomes that pair (a same-parity residual, handled in Step 4). If one active child of free type remains and $q = 1$ or $q \geq 3$, a helper port labelled $-x$ absorbs it. If $q = 2$, two ports labelled a and $-x - a$ absorb it with a token of rank two. If $q = 0$ then $|A| \geq 1$, so a neutral pair is undone and v goes to Step 4 with the pair as residual and the active child. A remaining child of a joined type is never absorbed by a helper (for the residue-one letter the helper would be $+x$, which is the head of the grandchild). Instead v receives the context $(h, \emptyset, p, c)$ with c the joined type.

Step 3a (forced-antipode children). The exact census shows exactly two contexts whose every family contains the antipode of its own head, the (1; one port) bottom and the two-input cell with head chain of length two and residual $\{1\}$. The first is folded into its parent at Step 3 in every case, as the child type L11, so it never becomes an owner, and the second is therefore the only forced-antipode child that occurs. Such a child c is absorbed as follows. If the parent is stationary with at least two ports, the port block containing the head of c absorbs it. If the parent is an owner whose context has an input-in-token family, that family absorbs it. Alternatively, and this is what happens for the contexts whose input-in-token search is infeasible, c is joined into the parent's cell as the child type A2, and the joint family is re-verified by the checker as a whole. The joined cell has a family that is not forced-antipode, so it is an ordinary child of the next parent and the chain stops after one step.

A two-input owner can have two forced-antipode children, and it then joins one of them and keeps the other pinned. In the joint the pinned child contributes one label whose output is that label itself, which is the role of a port, so the joint is searched as the same context with one more port and child type A2, and its family is required to carry the port in a token and to have no label identically minus the port. This is why we always join, and do not absorb, when both are possible: a cell that is joined into its parent is then a two-input cell with no forced-antipode children of its own, so the two grandchildren that the joint pins are ordinary, and no cell ever needs more than one token for an absorption.

The residue-one letter needs a word of its own, although it is not forced-antipode. With a direct parent edge and no port its own context has no family with a free head, so it is folded into its parent as the child type L1. With a subdivided parent edge, or with a port, it is an owner of its own. However, when its active child is forced-antipode the folding is not what we do: the forced-antipode child is joined into the letter instead, which gives the context $(h, \{1\}, p, A2)$, and that context has a family that is not forced-antipode, so the letter is an ordinary owner and its parent sees an ordinary free child. Therefore an L1 joint never carries a forced-antipode grandchild. (The opposite-parity bottom $(1, 2)$ was joined in an earlier version of the alphabet, although it has a family without the antipode of its head and is now an ordinary owner.)

Step 4 (nonempty residual). v is an owner with context $(h \bmod 6, R', p, c)$ where R' is R after the following repairs: a portless residual $\{1\}, \{2\}, \{4\}, \{6\}$ or $\{8\}$ with no active child retains one neutral unit, and a lone odd arm with a head chain likewise retains one unit (its portless cell has only rays). One unit exists in each case because v has at least two children. The number of recruited ports is $p = q$ when $q \leq 2$, otherwise $p \in \{1, 3\}$ with $q - p \neq 1$, so that the unrecruited ports form a block of size zero or at least two. Every tuple produced in this way is in the list, itself generated by the same rules.

Step 5 (root). The root has no head and its context is (root, R', p, c) with the same conventions, the root family containing the zero label. Exact search shows that a lone residual arm at the root needs two recruited ports, or none for odd residue and one for even residue. When the available ordinary children do not permit this, one neutral pair is retained, and when the root has a lone arm, no ports, no unit and $k \geq 2$ active children the exact root table decides between cancelling all active heads and keeping one or two active children. A root whose only children are one residual arm and one active child cannot occur, because branch vertices have three children.

Every owner is assigned exactly one context, every vertex of degree two lies in exactly one arm, head chain or unit, and therefore the cells and units are edge-disjoint and cover all subdivided edges. $\square$

The steps above are also the specification of our scheduler, and the abstract enumeration described in Section 3 is our check that the list of contexts generated from these steps is complete: the residual types, the port counts and the child types are finite, the head-chain and arm lengths enter only through their residues modulo six (with the raw lengths 1 and 2 distinguished), and every combination is exercised by at least one fragment.

B Proof of the realization lemma

Families, sinks and the choice rule

Every family carries the properties (E1)–(E5) of Section 4, which our checker verifies. Property (E6) holds for a family of every context that occurs except the 141 of Section 4, and for those the gate of Section 8 checks the finitely many adjacent pairs individually. For every owner

we fix a family before any label is chosen, preferring families with (E6). With this choice no non-input label of a parent is identically equal, as a form in the shared head, to a label of its child: the parent's only input-only label is the antipode of its input, and the child has no label equal to the antipode of its own head.

The tokens of a family are what (E2) leaves after the mirror pairs, but a token may contain the input x while the other token of the same family contains its antipode $-x$ (which (E6) allows), or two tokens may otherwise contain a label and its negative; such labels are already mirrored inside the cell (the input x is the head edge of the active child, $-x$ a label of the cell) and must not be negated again. We therefore work with *units*: take the multiset of the token labels of a family, remove every antipodal pair of forms among them, and call each zero-sum triple that remains, or the zero-sum set of four labels that remains when one pair straddled the two tokens, a unit; what remains has size 6 (two triples), 4 (a *quad*), 3 (a triple), 2 or 0 (nothing to place). The first token of a forced-antipode family, which contains the antipode of the head, keeps the treatment below; every other unit is *free*. The zero-sum triples that Section 3 uses to cancel an odd number of active heads at a vertex are free units as well. They are not closed under negation, so their negatives need sinks of their own; each of them consumes three active children, all of which are owners, and triples at different vertices consume disjoint sets of owners, which is what bounds their number in Lemma B.4. Free units are never cancelled against one another along the root path. Instead the negatives of each free unit are placed in the ordinary part of the tree, at a *sink*, and this is what closes the special support under negation. There are two kinds of sink. A stationary vertex whose ordinary block has size r takes negated units directly as long as the remainder of r after the units it holds is zero or at least two, as Section 6 requires (a single triple needs $r = 3$ or $r \geq 5$, a single quad $r = 4$ or $r \geq 6$); the units are zero-sum, so the block remains zero-sum. Otherwise units go to *gadgets* built from stationary vertices with two children: pairs of triples to the *absorbers* of Lemma B.1, quads to the four-vertex gadget of Lemma B.2. Every cell chooses all of its parameters numerically, in post-order, so that no symbolic parameter is ever live except in the two situations that already had their own treatment, the forced-antipode children and the two-input rows. This is a change of architecture with respect to an earlier version of this appendix, in which tokens were cancelled in pairs along the root path by solving $T_v = -T_A$ for the carrier of the resolving cell. That step divided by the determinant of a carrier block, and a chain of cells of the context `h0_3_p0_act` (a comb whose core vertices carry one arm each) has a one-dimensional space of solutions, so that the labels either collide or grow geometrically along the chain; Remark B.6 records this.

Lemma B.1 (five-point absorber). *Let v_0, v_1, v_2, v_3, v_4 be vertices with v_i the parent of v_{i+1} , none of them an owner, a helper or a vertex of an arm, head chain or unit, such that each v_i with $i \leq 3$ has a further child w_i that is stationary, v_4 has two such children w_4, w'_4 , and the remaining children of every v_i number zero or at least two. Let $T = (a, b, -a - b)$ and $U = (c, d, -c - d)$ be integer zero-sum triples. Give the edges the labels*

$$\begin{aligned} e(v_0v_1) &= a + b + d, & e(v_0w_0) &= -a - b - d, \\ e(v_1v_2) &= a, & e(v_1w_1) &= -c - d, \\ e(v_2v_3) &= b, & e(v_2w_2) &= d, \\ e(v_3v_4) &= 2a + b - c, & e(v_3w_3) &= c - 2a - b, \\ e(v_4w_4) &= c, & e(v_4w'_4) &= -a - b, \end{aligned}$$

and give the remaining children of every v_i a zero-sum block. Then v_0 and v_3 are stationary, v_1, v_2, v_4 are non-stationary with $s(v_1) = 2a + b - c$, $s(v_2) = a + b + d$ and $s(v_4) = a$, which are the labels of the parent edges of v_4, v_1, v_2 , so that the identity of Lemma 2.1 holds on the five vertices with the parent edge of v_0 untouched; the ten labels placed are $T \sqcup U \sqcup \{\pm(a + b + d)\} \sqcup \{\pm(2a + b - c)\}$; and as linear forms in (a, b, c, d) the ten labels are pairwise distinct

and nonzero, the only antipodal pairs among them being the two just named. Distinctness of the ten values is then a finite list of proper conditions on (a, b, c, d) , which the choice rule of Appendix B adds to Φ like any other; the lemma asserts it of the forms, not of every numerical T and U .

Proof. The sums are $s(v_0) = y + (a + b + d) - (a + b + d) = y$ for the parent label y of v_0 , $s(v_1) = (a + b + d) + a - c - d = 2a + b - c$, $s(v_2) = a + b + d$, $s(v_3) = b + (2a + b - c) + (c - 2a - b) = b$ and $s(v_4) = (2a + b - c) + c - a - b = a$. A zero-sum block at v_i does not change $s(v_i)$, and a stationary child w has $s(w) = e(v_i w)$, so a hanging label is produced by its own lower endpoint exactly as the label of a stationary child edge is. The list of labels is read off, and the distinctness of the ten forms and of their antipodal pairs is a finite check, which we also ran by script. $\square$

If two owners carry free tokens $-T$ and $-U$, the absorber places T and U and closes the special support under negation at the cost of two inverse pairs, and it imposes no equation on either owner: the four coordinates a, b, c, d are whatever the two cells chose. The three vertices v_1, v_2, v_4 join W . On a vertical path of stationary vertices with two children each a single token cannot be absorbed, since such a path places an even number of labels, and three such vertices do not suffice for two tokens with independent coordinates, because the three sums would have to permute the three path labels and the parent label, which forces either an antipodal pair inside $T \sqcup U$ or a dependence of the parent label on the tokens. Four vertices do not suffice either. With path labels g_1, g_2, g_3 below the parent label y and hanging labels p_0, p_1, p_2, p_3, p'_3 , the sum at the top vertex must be y (any other value would make y depend on the tokens or place $-y$), so the one inverse pair is $\{g_1, p_0\}$ and the six remaining labels are $T \sqcup U$; the sums at the next two vertices cannot equal their own path labels (that would force an antipodal pair inside $T \sqcup U$), so they are g_3 and g_1 in this order, which gives $g_1 + g_2 + p_1 = g_3$ and $g_2 + g_3 + p_2 = g_1$, hence $2g_2 = -(p_1 + p_2)$, and twice a label of $T \sqcup U$ is never the sum of two others when the coordinates are independent. Thus five vertices is the minimum for two tokens on a binary path.

Lemma B.2 (quad gadget). *Let v_0, v_1, v_2, v_3 be vertices with v_i the parent of v_{i+1} , eligible as in Lemma B.1, each v_i with $i \leq 2$ having a further stationary child w_i and v_3 two such children w_3, w'_3 . Let $Q = (q_1, q_2, q_3, q_4)$ be an integer zero-sum quadruple and f an integer. Give the edges the labels*

$$\begin{aligned} e(v_0 v_1) &= f - q_1 - q_2, & e(v_0 w_0) &= -(f - q_1 - q_2), \\ e(v_1 v_2) &= q_1, & e(v_1 w_1) &= q_2, \\ e(v_2 v_3) &= f, & e(v_2 w_2) &= -f, \\ e(v_3 w_3) &= q_3, & e(v_3 w'_3) &= q_4. \end{aligned}$$

Then v_0 is stationary, v_1, v_2, v_3 are non-stationary with sums f, q_1 and $f - q_1 - q_2$, the labels of the parent edges of v_3, v_2, v_1 respectively, so that the identity of Lemma 2.1 holds on the four vertices with the parent edge of v_0 untouched; the eight labels placed are $Q \sqcup \{\pm f\} \sqcup \{\pm(f - q_1 - q_2)\}$, and they are pairwise distinct and nonzero as forms in (q_1, q_2, q_3, f) .

Proof. $s(v_1) = (f - q_1 - q_2) + q_1 + q_2 = f$, $s(v_2) = q_1 + f - f = q_1$ and $s(v_3) = f + q_3 + q_4 = f - q_1 - q_2$; the rest is as in Lemma B.1. $\square$

The parameter f is ours, chosen when the gadget is filled, so the conditions that its four labels $\pm f, \pm(f - q_1 - q_2)$ avoid every placed value are proper affine conditions in f , and a quad gadget occupies the top four vertices of any five-vertex path of the kind Lemma B.4 provides.

Remark B.3 (a quad splits into two triples). The negatives $Q = (q_1, q_2, q_3, q_4)$ of a free quad need not go to a gadget or to a block that can hold four labels. They can be placed as the two triples $\{q_1, q_2, -q_1 - q_2\}$ and $\{q_3, q_4, -q_3 - q_4\}$ at two ordinary triple sinks. Both sets are

zero-sum, so both blocks stay zero-sum, and because Q is zero-sum the two labels introduced are $-q_1 - q_2$ and its negative, an inverse pair; the special support therefore stays closed under negation and the ledger gains nothing that needs a sink of its own. Each new label is a sum of two coordinates of Q , so the magnitude bound at the end of this appendix covers it, and each carries one further genericity condition, of the same kind as the conditions on f above. We therefore count every quad as *two* triples in Lemma B.4. This is what makes cap , which counts triples, the right capacity for an arbitrary mixture of triples and quads: a block of size three or five has capacity one but cannot hold a quad at all, so a capacity counted in units rather than in triples would not do.

Lemma B.4 (sinks exist). *Let t be the number of free units, a quad counted as two triples (Remark B.3). If $n \geq 10^6(D+1)$ then the direct capacities of the ordinary blocks, $\sum_u \text{cap}(r_u)$ with $\text{cap}(r) = \max\{k : r - 3k \in \{0\} \cup [2, \infty)\}$, together with the number of pairwise vertex-disjoint vertical paths of five stationary vertices whose ordinary blocks have size two or four, is at least t . (A path takes two triples, or one quad, which counts for two; a block of size r takes a quad in place of two triples whenever $r - 4 \in \{0\} \cup [2, \infty)$, and otherwise the quad is split as in Remark B.3. Counting a quad as two triples on either route is therefore conservative.)*

Proof. By Lemma 3.9 there are at most $2q \leq 2D$ owners, each with at most two units of its own, and a unit counts for at most two triples, so the units of the families contribute at most $8D$. A cancelling triple of Section 3 consumes three owners, and triples at different vertices consume disjoint sets of them, so there are at most $2D/3$ of those. Hence $t \leq 8D + 2D/3 \leq 9D$; helpers are at most one per owner. The skeleton, by which we mean the vertices of W , the arms with their leaves, the units, the head chains, the owners, the helpers, the helper ports (at most two per helper) and the recruited ports (at most three per owner), has at most $16D + 1$ vertices. Suppose the direct capacities sum to less than t . Then fewer than t blocks have positive capacity, and a block of capacity $k \geq 1$ has size at most $3k + 4$ (otherwise $k + 1$ would be admissible), so these blocks have fewer than $7t \leq 63D$ children in total. Every vertex outside the skeleton is a child in exactly one ordinary block, and blocks of size one do not exist, so all but $7t$ of the $n - 16D - 1$ non-skeleton vertices are children in blocks of size two or four. Call a vertex *usable* if it is neither an owner, nor a helper, nor on an arm, head chain or unit, and its block has size two or four; let U be the set of usable vertices and $m = |U|$. The parent of a child in a block of size two or four is usable or in the skeleton, and has at most four such children, so $n - 16D - 1 - 63D \leq 4(m + 16D + 1)$, i.e. $m \geq n/8$ once $n \geq 286D + 10$.

Consider the forest F induced on U . A vertex of U has at most four children in U , and the roots of F are children of skeleton vertices or of blocks with positive capacity, so F has at most $4(16D + 1) + 63D = 127D + 4$ components. Give each vertex of F its height, leaves of F having height 0. A vertex of height at most 4 lies either in a component of height at most 4, which has at most $1 + 4 + 16 + 64 + 256 = 341$ vertices, or below a child of height at most 4 of a vertex of height at least 5, and every vertex of height at least 5 has at most four such children, each with at most 341 vertices below it. Writing H for the number of vertices of height at least 5, we get $m \leq 1365H + 341(127D + 4)$, hence $H \geq m/2730$ once $m \geq 86614D + 2728$, which holds when $n \geq 7.0 \cdot 10^5(D+1)$. For a vertex u of height $h \geq 5$ with $h \equiv 0 \pmod{5}$ follow the path from u downward through children of maximal height; it visits heights $h, h-1, \dots, h-4 \geq 1$, five usable vertices; the first four have a child on the path and therefore, since their blocks have size two or four, a further child on which a gadget label can hang, and the last has height at least one, so its block of size two or four consists of children off the path, two of which take the last two labels. A hanging child may itself be the top vertex of another gadget; it is then stationary all the same, which is all that is required of it. Two such paths are disjoint, because from incomparable vertices they lie in disjoint subtrees, and from comparable vertices u above u' the heights satisfy $h(u) \geq h(u') + 5$, so the height ranges do not meet. Thus there are at least $H/5 \geq m/13650 \geq n/109200$ disjoint paths, which is at

least $t \leq 9D$ once $n \geq 982800D$. The three thresholds are implied by $n \geq 10^6(D + 1)$. $\square$

The constant is crude, and it is irrelevant beside A below. In our experiments the direct capacity alone covers the tokens on random sparse trees with $n \geq 20(D + 1)$, e.g. by a factor of at least twelve at $D \approx n/25$, and on a comb attached to a complete binary tree of depth d , where only absorbers can help, the greedy packing finds 2^{d-4} disjoint paths. Blocks at owners (their unrecruited ports) and at helpers (their further ports) count toward the direct capacity as well. A usable vertex with a block of size four gives its two further children an inverse pair, which does not change its sum.

Lemma B.5 (compatibility). *Send every free quad to a quad sink and pair the free triples arbitrarily, with at most one left over; a dependent pair of triples (two units of one cell, or of a cell and its active child) is given a role order for which none of the four absorber forms $\pm(a + b + d)$, $\pm(2a + b - c)$ is identically equal, over the joint parameters of the two cells, to a label of either cell, to the negative of one, to another absorber form or to zero. Such an order exists for every pair of catalogue families. Then, when the owners are processed in post-order with all parameters numeric, every condition that has to be avoided, including the distinctness of the negated units placed at the sinks from every other label, is a proper affine condition in the coordinate to which it is charged, and Lemma 3.9 applies.*

Proof. The labels of a family are homogeneous linear forms, pairwise distinct and nonzero by (E3). We first dispose of the negated units themselves. Let u be a label of a free unit of the cell v and consider the value $-u$ placed at the sink. It is not identically a label of v : the units were formed by removing every antipodal pair among the token labels, and a label outside the tokens is a mirror-pair member, whose negative is its partner and not $-u$. It is not identically a label of the active child of v , whose labels are forms in that child's parameters: the only parameter shared is the child's head x , and when both cells satisfy (E6) the child has no non-head label that is a multiple of x , while $u = -x$ would mean that x and $-x$ both occur among the tokens of v and were removed. It is not identically a label of the parent of v , which shares only the head y of v : under (E6) the parent's labels depending on y alone are $\pm y$, and $u = -y$ occurs only in the first token of a forced-antipode family, which is not free. For the finitely many pairs in which a cell violates the second clause of (E6) (a unit label that is a multiple of the input by a factor other than ± 1 , of which the catalogue has 30 in 28 contexts, with factors ± 2 , ± 3 , $\frac{1}{2}$, $\frac{3}{2}$; or a label that is such a multiple of the head, of which there are 478 in 166 contexts) the identity is checked directly, as part of the gate of Section 8, and the choice rule keeps the colliding combinations apart. Every other label of the tree is numeric when the sink is filled, so the remaining requirements on $-u$ are proper conditions on the parameters of v . The four labels of a quad gadget are affine in the fresh parameter f with slope ± 1 , so their conditions are proper in f . Consider now an absorber whose first source is already numeric and whose second source is the present cell with token $(c, d, -c - d)$. Each absorber form is a numeric constant plus one of $d, -c, -d, c$ (or, with the roles exchanged, plus $a + b, 2a + b$ and their negatives), so it has a nonzero coefficient in some parameter of the present cell. Its collision with a placed numeric label is therefore a proper condition. Its collision with a label ℓ of the present cell reads $d - \ell = \text{const}$, which is proper unless $d = \ell$ identically, and (E3) excludes that; collisions among the four absorber forms and the two tokens read $c \pm d = \text{const}$, $c = \text{const}$ or $d = \text{const}$, again proper because $c, d, -c - d$ are distinct nonzero labels. If the two sources are independent this is all. If they are dependent, the difference of two forms is a form over the joint parameters; it is proper for the cell processed second if it involves one of that cell's parameters, it is a proper condition on the cell processed first if it involves only that cell's parameters (and the pairing is known before any label is chosen, so the condition is recorded in time), and only an identical zero would be fatal. The finite check over every pair of a family with one input and a first token and a family with a head and a first token,

over every entry of every menu, 9,637,425 ordered pairs in all, finds a collision-free role order for each of them, in fact at least six of the thirty-six; the same check for two-input parents, second tokens and the two tokens of one cell is described in Section 8. $\square$

Without the finite check one would pair independent tokens only, which is possible for $t \geq 10$ because the dependency graph has maximum degree four and its complement then has minimum degree at least $t/2$; the check makes this unnecessary.

Remark B.6 (why the tokens are not cancelled along the root path). An earlier version of this appendix cancelled a free token against the first token on the path to the root by solving $T_v = -T_A$ for the carrier of the resolving cell. Three things go wrong with this, and we record them because each was found only after the argument had been written out. First, when the pending cell is the active child of the resolving cell, the child's head is the parent's input and appears on both sides of the equation, so the displayed solution is not a definition; the correct statement is two equations in three unknowns, whose relevant minors are mixed ones and not the determinant of a carrier block. Second, the division by that determinant was measured over the coordinates (y, x) with the family's denominator kept apart; measured on the lattice of parameter values at which the family's labels are integers, the token of `h0_3_p0_act` is a bijection onto the integer zero-sum triples and nothing is divided, whereas the tokens of `h1_2_p0_act` and `h4_1-3-3-3_p0_act` have image index two and three. Third, and decisively, the four contexts whose tokens depend on the head and the input alone have one free parameter per cell, so a chain of m of them, which our scheduler produces on a comb with one arm of length three at every core vertex, has a one-dimensional space of solutions once the tokens are paired, whatever the pairing: the heads either collide or grow by a factor of at least $5/2$ per pair, and our proof-faithful constructor records the failure on such a comb attached to a dense part. A two-level joint of two such cells does have a tokenless family with a free head (we verified this for all nine ordered pairs among the three small contexts), which would cut the chains as Lemma B.7 cuts the two-input chains. However, sending the tokens to sinks removes the pairing altogether, and with it the division, the chains and the live sets, so we use the sinks.

State

Owners are processed in post-order. Before any label is chosen the free tokens are paired and every token or pair is assigned a sink by Lemma B.4, and every dependent pair is given a collision-free role order by Lemma B.5. The state consists of the set `Placed` of numeric special labels, the set of *reserved* values (antipodes of heads that a parent will place later), the set `Open` $\subseteq$ `Placed` of placed labels whose antipode is not yet placed and is designated (the head of an active child, the three labels of a free token whose sink is not yet filled), and the set Φ of recorded forbidden conditions. The only symbolic labels are those of a forced-antipode child before its parent is processed, of a two-input row of residual $\{1\}$ before its reset, and the four labels of an absorber whose second source is the present cell; each is an affine form in the parameters of at most two adjacent cells.

A numeric value v is *admissible* for a label if $v \neq 0$, v is not placed or reserved, and $-v$ is not placed or reserved, unless $-v$ is open and v is its designated antipode (the forced $-x$ of an absorbing family, a helper, or a coordinate of a negated token when a sink is filled).

The invariant is: (a) numeric labels are pairwise distinct and nonzero, and no two are antipodal except designated pairs; (b) every symbolic label is a non-constant affine form in the parameters of the present cell or of its active child, and no two symbolic labels are identical or antipodal as forms; (c) Φ contains every condition under which a symbolic label would collide with a placed value or with another symbolic label; (d) $|\text{Placed}| + |\text{Reserved}| \leq |S| + \#\text{cells}$ and $|\Phi| \leq 3W(2|\text{Placed}| + 3W)$.

Processing an owner

Let v be the next owner with family F . Its input is the head label of the remaining active child, which is numeric, since every cell chooses its head numerically; the one exception is a child of the two-input residual- $\{1\}$ type, treated below.

No token. Choose the head and the internal parameters numerically, one coordinate at a time, in $M\mathbb{Z} \cap [-B, B]$, in a fixed order of the parameters. Property (E3) says that each label form is nonzero and that no two of them are equal, so each label form, and each difference of two label forms, has a well-defined *last* nonzero coordinate in that order. We charge every admissibility requirement and every reservation to the step at which the last nonzero coordinate of the form concerned is chosen; at that step the form is affine in the coordinate being chosen with nonzero slope, whereas at every later step it is already numeric. Each requirement therefore excludes at most one value, at exactly one step, and the reservation is $\{qy : q \in \Phi(F_{\text{parent}})\}$. We note that (E3) alone would not justify this coordinate by coordinate, since a label may well have zero coefficient in several parameters; it is the charging to the last nonzero coordinate that makes the count valid. The bound $B = M(2W(|\text{Placed}| + |\text{Reserved}| + 1) + W^2 + 1)$ therefore suffices.

Free token. Choose all parameters numerically exactly as in the previous case, with two additions to the forbidden set. If the token's sink is an absorber whose other source is already placed, the four absorber forms of Lemma B.1, with the other token's coordinates numeric and this token's coordinates as forms in the present parameters, are added to Φ , together with their negatives; by Lemma B.5 each is a proper affine condition, and there are at most eight of them. If the two sources are dependent and the present cell is the one processed first, the conditions of Lemma B.5 that involve only the present cell's parameters are added as well. When the last source of a sink is placed, the sink is filled: a direct sink receives the negated triple as three prescribed members of its block, and an absorber receives its ten labels, of which the six of $T \sqcup U$ are the designated antipodes of the two tokens and the four others are fresh; all become placed. The three labels of a token are open from the moment they are placed until the sink is filled.

Two-input rows. Treated identically with both inputs. The rows we call rigid need a word here, because their heads are written in Appendix A as fixed forms in their inputs, and so are their other labels. Our constructor carries them as templates outside the catalogue: the residue-two ray as $(-3t; -t, -2t, t; 2t)$ with input $x = 2t$ and head $-3x/2$, and the direct-edge two-input rows with residual $\{1\}$ and $\{2\}$ as $(-x_1 - x_2; -x_2, -x_1; x_1, x_2)$ and as the $\frac{1}{2}$ -form of Appendix A. (Solving $P' = O$ for the residue-two topology does give a two-dimensional space containing the template, but its second direction does not meet the signature of Section 4, so we do not use it, and the template's head is not a free parameter.) A rigid row has no free unit, and all of its labels are determined by its inputs; every requirement on such a label is charged to the step at which the last of those inputs is chosen, exactly as the coordinate-by-coordinate rule charges a label to its last coordinate, and the labels of a rigid row are reserved from that step on. One more thing has to be checked for the residue-two ray, whose non-head labels are $x/3, 2x/3, -x/3, -2x/3$ in terms of the head x that its parent receives as input: a parent family with a label that is one of these multiples of its input would collide identically. The choice rule therefore excludes such families above a residue-two ray; Section 8 reports the count of families and contexts concerned. For the row with residual $\{1\}$ an exhaustive enumeration of the bijections between the five labels and the five outputs of that topology leaves exactly one nondegenerate family, and its head is $-x_1 - x_2$. That row has no token, so a chain of such rows carries a symbolic head upward for as many levels as the chain is long. Our scheduler does produce arbitrarily long such chains, on a comb whose every core vertex has two active children and one arm of length one, so the length is $\Theta(D)$ in the worst case and not $O(1)$. What bounds it is Lemma B.7.

Lemma B.7 (the two-input reset). *Two consecutive residual- $\{1\}$ direct-edge two-input rows, joined into a single cell, have a family whose head is an independent parameter.*

Proof. The joint has nine labels, the heads g_0 and g_1 of the two rows, the two labels of each arm, the pinned input s_0 of the upper row and the two pinned inputs x_1, x_2 of the lower one, and nine outputs. Enumerating every bijection between them and solving the resulting linear system gives 14 families whose labels are distinct and nonzero at a generic point, and for 12 of them the space of solutions contains a direction along which g_0 moves while every input is held fixed. One such family, at an integer point, is

$$(g_0, g_1) = (14, -16), \quad (r_0 a_0 x_0, r_0 a_0 x_1) = (13, -29), \quad (r_1 a_0 x_0, r_1 a_0 x_1) = (-9, 22),$$

$$s_0 = 3, \quad (x_1, x_2) = (-3, 19),$$

whose solution space has dimension five and whose head-free direction is g_0 itself, so g_0 may be moved on its own. We verified $P = O$ and the distinctness of the nine labels at that point directly. (This family is the one with the fewest moving labels once the inputs are pinned; the constructor uses a different one of the twelve, see below.) $\square$

A chain of such rows is therefore cut into joints of two levels, each of which has a free head, and a chain of odd length leaves one single row at its top whose parent is not one of these rows. The symbolic labels are confined to at most two cells and one further row, where a cell may now be a joint of width nine, so the bound is a constant and the boxes and the magnitudes are again linear in $D + 1$. Of the 12 head-free families, seven have the further property that all six labels placed by the joint still move once its three inputs are pinned, and our constructor uses one of these; for the single row that an odd chain leaves at its top, whose labels are determined by its inputs, the requirements are charged to the step at which the last of those inputs is chosen, exactly as in the coordinate-by-coordinate rule above. Our scheduler reports these joints (field `reset_joints`) and the constructor of Section 8 realizes them as single cells.

Stationary vertices. A helper places $-x$, a two-port absorption places a fresh numeric a and $-x - a$, and neutral units are instantiated with fresh numeric parameters. A stationary vertex that is a sink is filled as described under *Free token*, and its remaining children, if any, are left to the completion as a zero-sum block of size at least two.

Two-token cells. The units of the cell (two triples, or one quad when the tokens share an antipodal pair) are sent to sinks like any other; they are dependent on each other and on the units of the parent and of the active children, which Lemma B.5 allows.

Forced-antipode children. The token $\{-x, a, x - a\}$ of a forced-antipode child contains the antipode of the child's own head edge x , whose negative is that edge itself, so it cannot be sent to a sink. It stays symbolic with its carrier live until the parent is processed. The parent's port block or input-in-token family supplies the token $\{x, b, -x - b\}$, and the single equation $a = -b$ is solved for the child's carrier (its token map has rank two on the carrier, so the equation is non-degenerate, and it involves no division since b is a single label). The second token of a forced-antipode family, when there is one (the families of the one forced-antipode context that occurs all carry two, the second being $\{x_1, x_2, -x_1 - x_2\}$ on the two inputs), is a free unit and goes to a sink, as is a second token of an input-in-token family of the parent.

Root. The root family contains the zero label; its free tokens go to sinks like any other. If the number of free tokens is odd, one token is left over by Lemma B.5; its negatives are handed to an odd ordinary block of size at least three, which exists because the special support then has odd size, so the ordinary block sizes have odd sum, and no block has size one.

Verification of the invariant and of the bounds

(a) follows from admissibility; (b) from (E3) and the choice rule for the forced-antipode and two-input cases and from Lemma B.5 for the absorber forms; (c) by construction; (d) because every owner adds at most W placed labels and at most $|\Phi(F)| \leq 10$ reservations, the bound 10 being computed from the catalogue as in Lemma 3.9, every absorber adds four placed labels and a direct sink none, and every absorber adds at most eight conditions to Φ . Every box has side $O(|\text{Placed}| + |\text{Reserved}| + |\Phi|) = O(|S|)$, every label is a bounded-coefficient combination of at most seven parameters, and $|S| \leq 12D + 3 + 2t \leq 30D + 3$ by Lemma 3.9 and $t \leq 9D$, so all magnitudes are at most $A(D + 1)$ for an absolute A . Explicitly, with $W = 29$ and $M = 24$, and with $|\text{Placed}| + |\text{Reserved}|$ together with the absorber conditions at most $(30 + 2 + 210 + 8)(D + 1) = 60(D + 1)$, the coordinate boxes have side at most $M(2W(60(D + 1) + 1) + W^2 + 1) = 83,520(D + 1) + 21,600$. A label of a cell is a sum of at most 7 terms with coefficients at most 64 over a denominator that only decreases the magnitude, so it has magnitude at most $7 \cdot 64 \cdot (83,520(D + 1) + 21,600) = 37,416,960(D + 1) + 9,676,800$, and an absorber label is a sum of at most four token coordinates, so every special label has magnitude at most $4(37,416,960 + 9,676,800)(D + 1) \leq 1.9 \cdot 10^8(D + 1)$. This is the constant A of Lemma 5.1; no lattice or coefficient enlargement occurs, since nothing is ever divided. At the end every special label other than a possible single leftover triple has its antipode placed, so the special support is closed under negation up to that triple. $\square$

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Table 1: The neutral blocks of Lemma 3.7. The labels are read as multiples of a fresh parameter t , the ports first and then the arms in the order listed, each arm from the owner outwards. The equal pairs are the closed form $\pm(-2)^i$; the others come from an exhaustive search and are re-verified from the labels alone.

ports	arms	token	labels
0	1, 1		1, -2, -1, 2
0	1, 5		4, -2, -4, 3, 1, 2, -1, -3
0	1, 6	yes	(-2, 3), (3, -4), (2, -3), (-1, 1), (-3, 5), (1, -2), (1, -1), (-4, 6), (3, -5)
0	1, 7		1, -2, -1, 2, 3, -8, 5, -3, -5, 8
0	2, 2		1, -2, 4, -1, 2, -4
0	3, 2	yes	(3, -2), (-6, 3), (3, -1), (0, -1), (-3, 2), (-3, 1), (6, -2)
0	2, 4		6, 2, -8, -6, 4, -2, 8, -4
0	2, 5	yes	(-3, 4), (0, 1), (3, -5), (3, -4), (-3, 5), (1, -2), (-2, 3), (-1, 1), (2, -3)
0	2, 6		5, -2, 4, -5, 3, 2, -3, -1, -4, 1
0	2, 8		8, -7, -1, -8, 3, -5, -2, 7, 1, 2, 5, -3
0	3, 3		1, -2, 4, -8, -1, 2, -4, 8
0	3, 4	yes	(2, 0), (-1, -6), (0, 6), (-2, -6), (-2, 0), (1, -6), (1, 6), (-1, 0), (2, 6)
0	3, 5		5, 1, 3, -6, -5, 4, -3, 6, -1, -4
0	3, 6	yes	(6, -5), (0, -1), (0, 2), (3, -3), (-6, 5), (6, -6), (0, 1), (-6, 4), (3, -1), (-3, 3), (-3, 1)
0	3, 7		1, 5, -6, -2, -1, 3, -8, 2, -5, 6, -3, 8
0	4, 4		1, -2, 4, -8, 16, -1, 2, -4, 8, -16
0	5, 4	yes	(6, -5), (0, 3), (0, -2), (0, 1), (0, -4), (0, 2), (-6, 5), (6, -2), (0, -3), (0, -1), (-6, 6)
0	4, 6		2, -1, 4, -7, 5, -2, -3, -4, 3, 1, -5, 7
0	4, 8		4, 1, -6, 2, -8, -4, 5, -7, 6, -2, -5, 7, -1, 8
0	5, 5		1, -2, 4, -8, 16, -32, -1, 2, -4, 8, -16, 32
0	5, 6	yes	(5, -4), (-5, 5), (0, -2), (-5, 3), (5, -1), (0, -3), (-5, 4), (0, 1), (0, 2), (5, -3), (-5, 1), (0, 3), (5, -6)
0	5, 7		2, -3, 4, -1, 6, -8, -2, -4, 8, -6, 3, 5, 1, -5
0	6, 6		1, -2, 4, -8, 16, -32, 64, -1, 2, -4, 8, -16, 32, -64
0	6, 8		1, 6, -7, 8, -3, 7, -5, -1, -6, 3, 5, -2, -4, 2, 4, -8
0	7, 7		1, -2, 4, -8, 16, -32, 64, -128, -1, 2, -4, 8, -16, 32, -64, 128
0	8, 8		1, -2, 4, -8, 16, -32, 64, -128, 256, -1, 2, -4, 8, -16, 32, -64, 128, -256
1	1, 2		3, -1, 2, -2, 1, -3
1	1, 4		6, -8, 4, 2, -4, -2, -6, 8
1	1, 6		2, 1, 4, -3, -1, -2, 5, -4, 3, -5
1	1, 8		1, -3, 8, 2, -5, 3, -4, -1, 5, -2, 4, -8
1	3, 2		3, -2, 1, -3, 5, -1, 2, -5
1	5, 2		3, -4, 2, -1, -2, -3, 5, 1, -5, 4
1	7, 2		5, -3, 2, -1, -4, 1, 3, -5, 8, -2, 4, -8
1	3, 4		5, -2, -1, -4, 2, -3, 4, -5, 1, 3
1	3, 6		2, 3, -2, 5, -8, -5, 4, 1, -3, -1, -4, 8
1	3, 8		4, 3, -5, -2, 7, -7, 6, -3, 5, 1, -4, -1, 2, -6
1	4		3, -3, 1, -2, -1, 2
1	5, 4		7, -5, 3, 5, -8, 2, -7, -2, -6, 8, -3, 6
1	7, 4		8, -2, 5, -1, -5, 2, 3, -4, 6, -6, 1, -3, 4, -8
1	5, 6		1, 2, 3, 5, -6, 4, -8, -3, -2, -1, -5, 8, -4, 6
1	5, 8		6, 1, 4, -8, 2, -5, 8, -7, 5, -6, -2, 3, -1, -4, -3, 7
1	6		5, -5, 3, -1, 2, 1, -2, -3
1	7, 6		7, 1, 4, -5, 3, -1, -7, 2, -6, -8, 5, -4, -3, 6, -2, 8
1	8		7, -7, 3, 1, -4, -3, 2, -1, 4, -2
2	—		1, -1
2	5		5, -3, -2, 1, 2, -1, 3, -5
2	7		3, -2, -1, 6, -5, 2, -3, 5, 1, -6

Table 2: The certificate table, by kind of context: the contexts that occur (Lemma 3.8), those for which the catalogue carries a family, the number of families, the contexts with a family satisfying (E6), and the contexts whose chosen family carries a token. The residue-two rigid row is a constructor template and not a symbolic family, so it is counted as covered in the second column and in none of the others.

context kind	contexts	with a family	families	(E6)	token
bottom cells	954	954	1,563	913	493
continuing cells, free child	419	419	1,783	417	406
continuing cells, joined child	876	876	1,807	810	440
two-input cells	34	34	213	9	33
root cells	507	507	1,332	499	352
total	2,790	2,790	6,698	2,648	1,724